

Nucleic Acid Structure and Gene Expression

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KEY CONCEPTS

- The great bulk of eukaryotic genetic information is stored in the DNA found in the nucleus. A tiny amount is also stored in mitochondrial and chloroplast DNA.
- DNA molecules are polymers of nucleotide repeat units that consist of one of four types of nitrogenous base, plus a sugar, plus a phosphate.
- The backbone of any DNA molecule is a sugar–phosphate polymer, but it is the sequence of the bases attached to the sugars that determines the identity and genetic function of any DNA sequence.
- DNA normally occurs as a double helix, comprising two strands that are held together by hydrogen bonds between pairs of complementary nitrogenous bases.
- Transmission of genetic information from cell to cell is normally achieved by copying the complementary DNA molecules that are then shared equally between two daughter cells.
- Genes are discrete segments of DNA that are used as a template to synthesize a functional complementary RNA molecule.
- Most genes make an RNA that will serve as a template for making a polypeptide.
- Various genes make RNA molecules that do not encode polypeptide. Such noncoding RNA often helps regulate the expression of other genes.
- Like DNA, RNA molecules are polymers of nucleotide repeat units that consist of one of four types of nitrogenous base (three of these are the same as in DNA), plus a slightly different sugar, plus a phosphate.
- Unlike DNA, RNA molecules are usually single-stranded.
- To become functional, newly synthesized RNA must undergo a series of maturation steps such as excising unwanted intervening sequences and chemical modification of certain bases.
- Polypeptide synthesis occurs at ribosomes, either in the cytoplasm or inside mitochondria and chloroplasts.
- The sequential information encoded in the RNA is interpreted at the ribosome via a triplet genetic code, determining the basic structure of the polypeptide.
- Polypeptides often undergo a variety of chemical modifications.
- Proteins display extraordinary structural and functional diversity.

1.1 DNA, RNA, AND POLYPEPTIDES

Molecular genetics is primarily concerned with the inter-relationship between two nucleic acids, DNA and RNA, and how these are used to synthesize polypeptides, the basic component of all proteins. RNA may have been the hereditary material at a very early stage of evolution, but now, except in certain viruses, it no longer serves this role. Genetic information is instead stored in more chemically stable DNA molecules that can be copied faithfully and transmitted to daughter cells.

Nucleic acids were originally isolated from the nuclei of white blood cells, but are found in all cells and in viruses. In eukaryotes, DNA molecules are found mainly in the chromosomes of the nucleus, but each mitochondrion also has a small DNA molecule, as do the chloroplasts of plant cells.

A **gene** is a part of a DNA molecule that serves as a template for making a functionally important RNA molecule. In simple organisms such as bacteria, the DNA is packed with genes (typically at least several hundred up to a few thousand different genes). In eukaryotes, the small DNA molecules of the mitochondrion or chloroplast contain a few genes (tens up to hundreds) but the nucleus often contains thousands of genes, and complex eukaryotes typically have tens of thousands. In the latter case, however, much of the DNA consists of repetitive sequences whose functions are not easily identified. Some of the repetitive DNA sequences support essential chromosomal functions, but there are also many defective copies of functional genes.

There are many different types of RNA molecule but they can be divided into two broad classes. In one class, each RNA molecule contains a **coding RNA** sequence that can be decoded to generate a corresponding polypeptide sequence. Because this class of RNA carries genetic information from DNA to the protein synthesis machinery, it is described as **messenger RNA (mRNA)**. Messenger RNA made in the nucleus needs to be exported to the cytoplasm to make proteins, but the messenger RNA synthesized in mitochondria and chloroplasts is used to make proteins within these organelles. Most gene expression is ultimately dedicated to making polypeptides, so proteins represent the major functional endpoint of the information stored in DNA.

The other RNA class is **noncoding RNA**. Such molecules do not serve as templates for making polypeptides. Instead they are often involved in assisting the expression of other genes, sometimes in a fairly general way and sometimes by regulating the expression of a small set of target genes. These regulatory processes may involve catalytic RNA molecules (ribozymes).

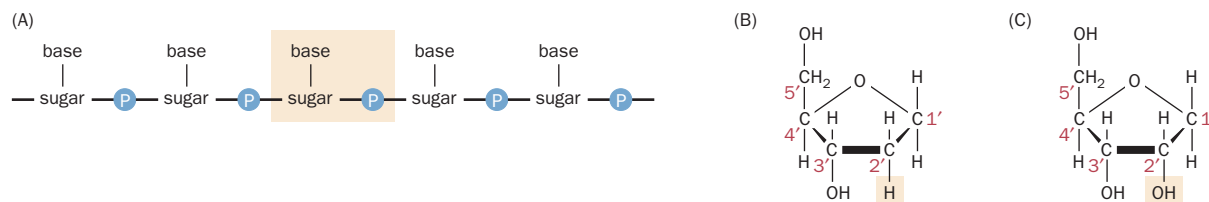
Most genetic information flows in the sequence DNA → RNA → polypeptide

Genetic information generally flows in a one-way direction: DNA is decoded to make RNA, and then RNA is used to make polypeptides that subsequently form proteins. Because of its universality, this flow of genetic information has been described as the central dogma of molecular biology. Two processes are essential in all cellular organisms:

- **transcription**, by which DNA is used by an RNA polymerase as a template for synthesizing one of many different types of RNA;
- **translation**, by which mRNA is decoded to make polypeptides at ribosomes, which are large RNA–protein complexes found in the cytoplasm, and also in mitochondria and chloroplasts.

Genetic information is encoded in the linear sequence of nucleotides in DNA and is decoded in groups of three nucleotides at a time (triplets) to give a linear sequence of nucleotides in RNA. This is in turn decoded in groups of three nucleotides (codons) to generate a linear sequence of amino acids in the polypeptide product.

Eukaryotic cells, including mammalian cells, contain nonviral chromosomal DNA sequences, such as members of the mammalian LINE-1 repetitive DNA family, that encode cellular reverse transcriptases, which can produce DNA sequences from an RNA template. The central dogma of unidirectional flow of genetic information in cells is therefore not strictly valid.



Nucleic acids and polypeptides are linear sequences of simple repeat units

Nucleic acids

DNA and RNA have very similar structures. Both are large polymers with long linear backbones of alternating residues of a phosphate and a five-carbon sugar. Attached to each sugar residue is a nitrogenous base (Figure 1.1A). The sugars in DNA and RNA differ, in either lacking or possessing, respectively, an -OH group at their 2'-carbon positions (Figure 1.1B, C). In deoxyribonucleic acid (DNA), the sugar is deoxyribose; in ribonucleic acid (RNA), the sugar is ribose.

Unlike the sugar and phosphate residues, the bases of a nucleic acid molecule vary. The sequence of bases identifies the nucleic acid and determines its function. Four types of base are commonly found in DNA: adenine (A), cytosine (C), guanine (G), and thymine (T). RNA also has four major types of base. Three of them (adenine, cytosine, and guanine) also occur in DNA, but in RNA uracil (U) replaces thymine (Figure 1.2A).

Figure 1.1 Repeat units in nucleic acids.

(A) The linear backbone of nucleic acids consists of alternating phosphate and sugar residues. Attached to each sugar is a base. The basic repeat unit (pale peach shading) consists of a base + sugar + phosphate = a nucleotide. The sugar has five carbon atoms numbered 1' to 5'. (B) In DNA, the sugar is deoxyribose. (C) In RNA, the sugar is ribose, which differs from deoxyribose in having a hydroxyl (OH) group attached to carbon 2'.

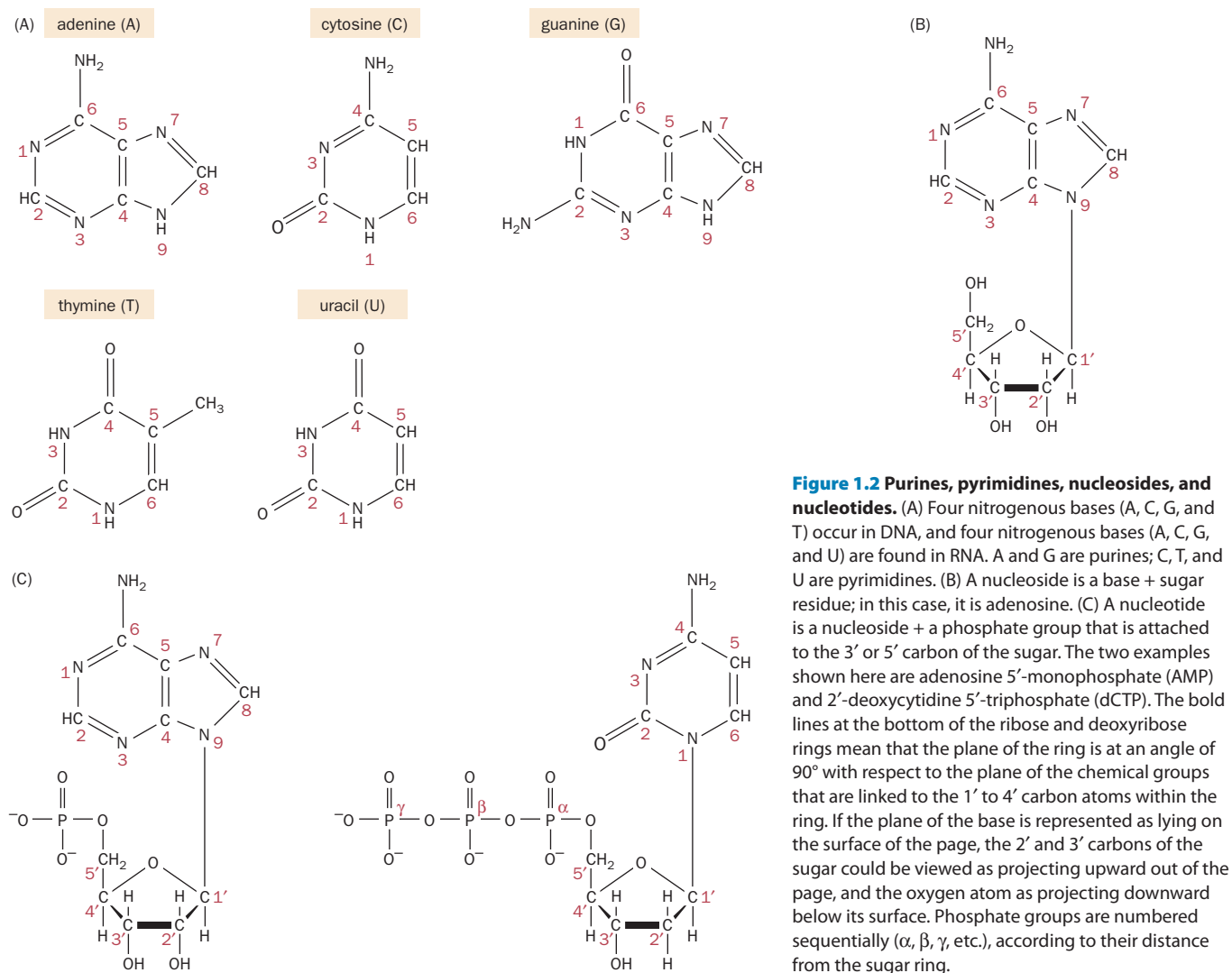


Figure 1.2 Purines, pyrimidines, nucleosides, and nucleotides.

(A) Four nitrogenous bases (A, C, G, and T) occur in DNA, and four nitrogenous bases (A, C, G, and U) are found in RNA. A and G are purines; C, T, and U are pyrimidines. (B) A nucleoside is a base + sugar residue; in this case, it is adenosine. (C) A nucleotide is a nucleoside + a phosphate group that is attached to the 3' or 5' carbon of the sugar. The two examples shown here are adenosine 5'-monophosphate (AMP) and 2'-deoxycytidine 5'-triphosphate (dCTP). The bold lines at the bottom of the ribose and deoxyribose rings mean that the plane of the ring is at an angle of 90° with respect to the plane of the chemical groups that are linked to the 1' to 4' carbon atoms within the ring. If the plane of the base is represented as lying on the surface of the page, the 2' and 3' carbons of the sugar could be viewed as projecting upward out of the page, and the oxygen atom as projecting downward below its surface. Phosphate groups are numbered sequentially (α , β , γ , etc.), according to their distance from the sugar ring.

TABLE 1.1 NOMENCLATURE FOR BASES, NUCLEOSIDES, AND NUCLEOTIDES

Base	Nucleoside = base + sugar		Nucleotide = nucleoside + phosphate(s)		
	Ribose	Deoxyribose	Monophosphate	Diphosphate	Triphosphate
Purine					
Adenine	adenosine		adenosine monophosphate (AMP) ^a	adenosine diphosphate (ADP)	adenosine triphosphate (ATP)
		deoxyadenosine	deoxyadenosine monophosphate (dAMP) ^b	deoxyadenosine diphosphate (dADP)	deoxyadenosine triphosphate (dATP)
Guanine	guanosine		guanosine monophosphate (GMP) ^a	guanosine diphosphate (GDP)	guanosine triphosphate (GTP)
		deoxyguanosine	deoxyguanosine monophosphate (dGMP)	deoxyguanosine diphosphate (dGDP)	deoxyguanosine triphosphate (dGTP)
Pyrimidine					
Cytosine	cytidine		cytidine monophosphate (CMP) ^a	cytidine diphosphate (CDP)	cytidine triphosphate (CTP)
		deoxycytidine	deoxycytidine monophosphate (dCMP)	deoxycytidine diphosphate (dCDP)	deoxycytidine triphosphate (dCTP)
Thymine	thymidine		thymidine monophosphate (TMP) ^a	thymidine diphosphate (TDP)	thymidine triphosphate (TTP)
		deoxythymidine	deoxythymidine monophosphate (dTMP)	deoxythymidine diphosphate (dTDP)	deoxythymidine triphosphate (dTTP)
Uracil	uridine		uridine monophosphate (UMP) ^a	uridine diphosphate (UDP)	uridine triphosphate (UTP)
		deoxyuridine	deoxyuridine monophosphate (dUMP)	deoxyuridine diphosphate (dUDP)	deoxyuridine triphosphate (dUTP)

^aNucleoside monophosphates are alternatively named as follows: AMP, adenylate; GMP, guanylate; CMP, cytidylate; TMP, thymidylate; UMP, uridylate.

^bWhere the sugar is ribose, the nucleotide is AMP; where the sugar is deoxyribose, the nucleotide is dAMP. This pattern applies throughout the table. Note that TMP, TDP, and TTP are not normally found in cells.

Bases consist of heterocyclic rings of carbon and nitrogen atoms, and can be divided into two classes: **purines** (A and G), which have two interlocked rings, and **pyrimidines** (C, T, and U), which have a single ring. In nucleic acids, each base is attached to carbon 1' (one prime) of the sugar; a sugar with an attached base is called a **nucleoside** (Figure 1.2B). A nucleoside with a phosphate group attached at the 5' or 3' carbon of the sugar is the basic repeat unit of a DNA strand and is called a **nucleotide** (Figure 1.2C and [Table 1.1](#)).

Polypeptides

Proteins are composed of one or more **polypeptide** molecules that may be modified by the addition of carbohydrate side chains or other chemical groups. Like DNA and RNA, polypeptide molecules are polymers that are a linear sequence of repeating units. The basic repeat unit is called an **amino acid**. An amino acid has a positively charged amino group ($-\text{NH}_2$) and a negatively charged carboxylic acid (carboxyl) group ($-\text{COOH}$). These are connected by a central α -carbon atom that also bears an identifying side chain that determines the chemical nature of the amino acid. Polypeptides are formed by a condensation reaction between the amino group of one amino acid and the carboxyl group of the next, to form a repeating backbone, where the side chain (called an R-group) can differ from one amino acid to another ([Figure 1.3](#)).

The 20 different common amino acids can be categorized according to their side chains:

- basic amino acids ([Figure 1.4A](#)) carry a side chain with a net positive charge at physiological pH;

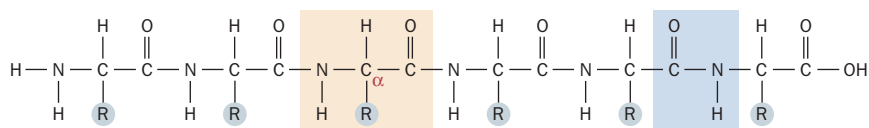


Figure 1.3 The basic repeat structure of polypeptides. A polypeptide is a polymer consisting of amino acid repeat units (pale peach shading). Amino acids have the general formula $\text{H}_2\text{N}-\text{CH}(\text{R})-\text{COOH}$, where R is the side chain, $\text{H}_2\text{N}-$ is the amino group, and $-\text{COOH}$ is the carboxyl group. The central α carbon carries all three groups, in each amino acid. The blue shading illustrates one of the peptide bonds that link adjacent amino acids.

- acidic amino acids (Figure 1.4B) carry a side chain with a net negative charge at physiological pH;
- uncharged polar amino acids (Figure 1.4C) are electrically neutral overall, although their side chains carry polar electrical groups with fractional electrical charges (denoted as $\delta+$ or $\delta-$);
- nonpolar neutral amino acids (Figure 1.4D) are hydrophobic (repelling water), often interacting with one another and with other hydrophobic groups.

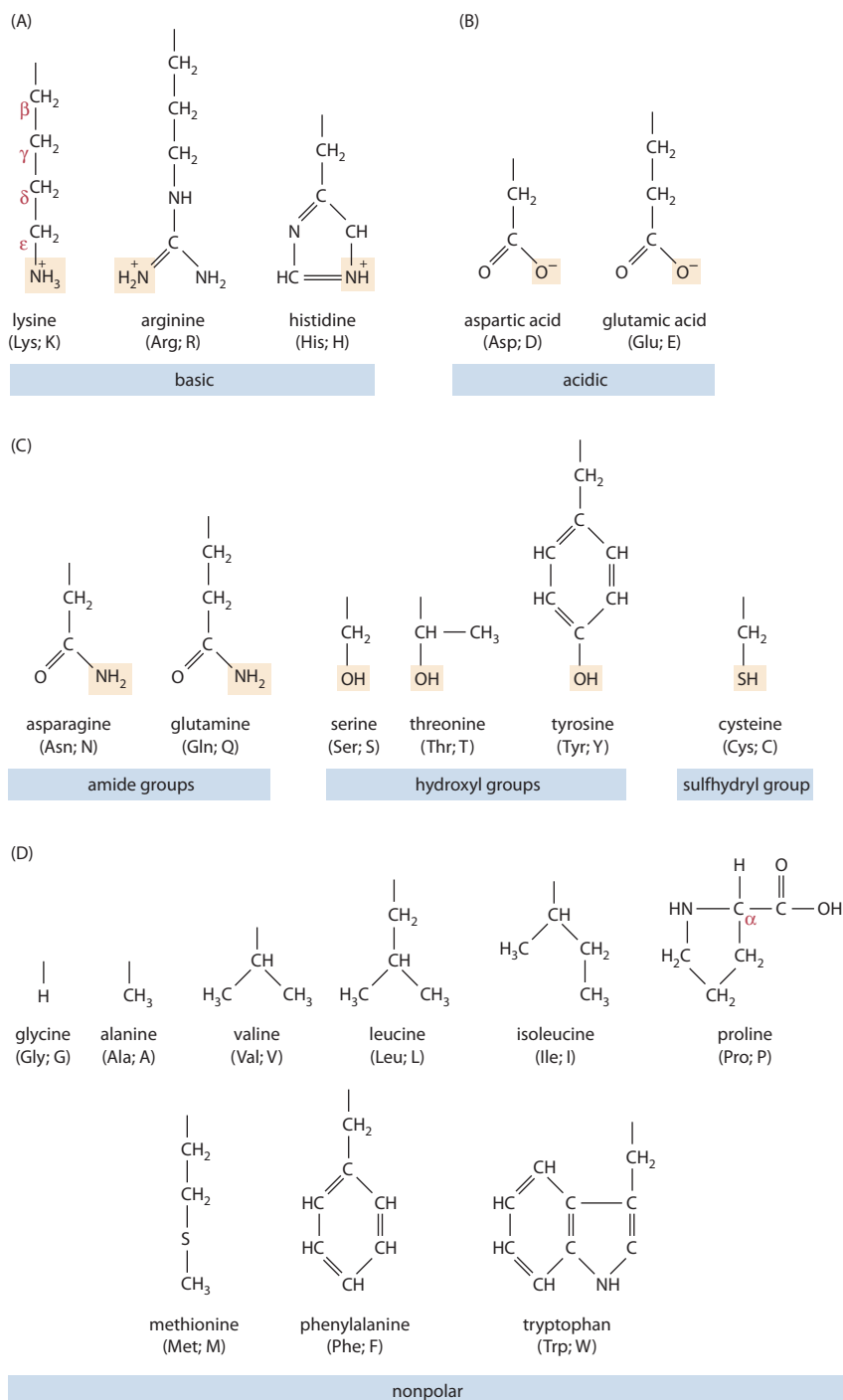


Figure 1.4 R groups of the 20 common amino acids, grouped according to chemical class. There are 11 polar amino acids, divided into three classes: (A) basic amino acids (positively charged); (B) acidic amino acids (negatively charged); and (C) uncharged polar amino acids bearing three different types of chemical group. Polar chemical groups are highlighted. (D) In addition, a fourth class is composed of nine nonpolar neutral amino acids. Amino acids within each class are chemically very similar. Side-chain carbon atoms are numbered from the central α -carbon atom (see the lysine side chain). In proline, the R group's side chain connects to the amino acid's $-\text{NH}_2$ group as well as to its central α -carbon atom.

TABLE 1.2 WEAK NONCOVALENT BONDING BONDS AND FORCES

Type of bond	Nature of bond
Hydrogen	Hydrogen bonds form when a hydrogen atom interacts with electron-attracting atoms, usually oxygen or nitrogen atoms
Ionic	Ionic interactions occur between charged groups. They can be very strong in crystals, but in an aqueous environment the charged groups are shielded by both water molecules and ions in solution and so are quite weak. Nevertheless, they can be very important in biological function, as in enzyme–substrate recognition
Van der Waals forces	Any two atoms in close proximity show a weak attractive bonding interaction (van der Waals attraction) as a result of their fluctuating electrical charges. When atoms become extremely close, they repel each other very strongly (van der Waals repulsion). Although individual van der Waals attractions are very weak, the cumulative effect of many such attractive forces can be important when there is a very good fit between the surfaces of two macromolecules
Hydrophobic forces	Water is a polar molecule. Hydrophobic molecules or chemical groups in an aqueous environment tend to cluster. This minimizes their disruptive effects on the complex network of hydrogen bonds between water molecules. Hydrophobic groups are said to be held together by hydrophobic bonds, although the basis of their attraction is their common repulsion by water molecules

In general, polar amino acids are hydrophilic, and nonpolar amino acids are hydrophobic. Glycine, with its very small side chain, and cysteine (whose –SH group is not as polar as an –OH group) occupy intermediate positions on the hydrophilic–hydrophobic scale.

As described below, the side chains can be modified by the addition of various chemical groups or sugar chains.

The type of chemical bonding determines stability and function

The stability of nucleic acid and protein polymers is primarily dependent on strong covalent bonds between the atoms of their linear backbones. In addition to covalent bonds, weak noncovalent bonds ([Table 1.2](#)) are important both between and within nucleic acids or protein molecules ([Box 1.1](#)). Individual noncovalent bonds are typically more than 10 times weaker than individual covalent bonds.

The structure of water is particularly complex, with a rapidly fluctuating network of noncovalent bonding occurring between water molecules. The predominant force in this structure is the **hydrogen bond**, a weak electrostatic bond between fractionally positive hydrogen atoms and fractionally negative atoms (oxygen atoms, in the case of water molecules).

BOX 1.1 THE IMPORTANCE OF HYDROGEN BONDING IN NUCLEIC ACIDS AND PROTEINS

Intermolecular hydrogen bonding in nucleic acids

This is important in permitting the formation of the following double-stranded nucleic acids:

- **Double-stranded DNA.** The stability of the double helix is maintained by hydrogen bonding between A–T and C–G base pairs. The individual hydrogen bonds are weak, but in eukaryotic cells the two strands of a DNA helix are held together by between tens of thousands and hundreds of millions of hydrogen bonds.
- **DNA–RNA duplexes.** Hydrogen bonds form naturally between DNA and RNA during transcription, but the base pairing is transient because the RNA migrates away from DNA as it matures.
- **Double-stranded RNA.** This occurs stably in the genomes of some viruses. It also arises transiently in cells during gene expression. For example, during RNA splicing, small nuclear RNA molecules bind to complementary sequences in pre-mRNA, and mRNA

codons bind to tRNA during translation. Many regulatory RNAs, such as microRNAs, control the expression of selected target genes by base pairing to complementary sequences at the RNA level.

Intramolecular hydrogen bonding in nucleic acids

This is particularly prevalent in RNA molecules. Intramolecular base pairing can form hairpins that may be crucially important to the structure of some RNAs such as rRNA and tRNA (see [Figure 1.9](#)), and as targets for gene regulation.

Intramolecular hydrogen bonding in proteins

Several characteristic elements of protein secondary structure, such as α -helices and β -pleated sheets, arise because of hydrogen bonding between side chains of different amino acids on the same polypeptide chain.

Charged molecules are highly soluble in water. Because of the phosphate groups in their component nucleotides, both DNA and RNA are negatively charged polyanions. Depending on their amino acid composition, proteins may be electrically neutral, or they may carry a net positive charge (basic protein) or a net negative charge (acidic protein). All of these molecules can form multiple interactions with the water during their solubilization. Even electrically neutral proteins are readily soluble, if they contain sufficient charged or neutral polar amino acids. In contrast, membrane-bound proteins with many hydrophobic amino acids are thermodynamically more stable in a hydrophobic environment.

Although individually weak, the numerous noncovalent bonds acting together make large contributions to the stability of the **conformation** (structure) of these molecules and are important for specifying the shape of a macromolecule. Covalent bonds are comparatively stable, so a high input of energy is needed to break them. Noncovalent bonds, however, are constantly being made and broken at physiological temperatures (see Box 1.1).

1.2 NUCLEIC ACID STRUCTURE AND DNA REPLICATION

DNA and RNA structure

DNA and RNA molecules have linear backbones consisting of alternating sugar residues and phosphate groups. The sugar residues are linked by **3', 5'-phosphodiester bonds**, in which a phosphate group links the 3' carbon atom of one sugar to the 5' carbon atom of the next sugar in the sugar-phosphate backbone (Figure 1.5).

Although certain viral genomes are composed of single-stranded DNA, cellular DNA forms a **double helix**: two strands of DNA are held together by hydrogen bonds to form a duplex. Hydrogen bonding occurs between the laterally opposed **complementary base pairs** on the two strands of the DNA duplex. Such base pairs form according to Watson-Crick rules: A pairs with T, while G pairs with C (Figure 1.6).

Because of base pairing, the base composition of DNA is not random: the amount of A equals that of T, and the amount of G equals that of C. The base composition of DNA can therefore be specified by quoting the percentage of GC (= percentage of G + percentage of C) in its composition. For example, DNA with 42% GC has the following base composition: G, 21%; C, 21%; A, 29%; T, 29%.

The two strands of a DNA double helix curve around each other to produce a minor groove and a major groove in the double helix, where the distance occupied by a single complete turn of the helix (its pitch) is 3.6 nm (Figure 1.7). DNA can adopt different types of helical structure. Under physiological conditions, most DNA in bacterial or eukaryotic cells adopts the B form, which is a right-handed helix (it spirals in a clockwise direction away from the observer) and has 10 base pairs per turn. Rarer forms are A-DNA (right-handed helix with 11 base pairs per turn) and Z-DNA (a left-handed helix with 12 base pairs per turn).

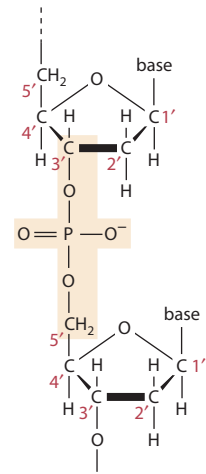


Figure 1.5 A 3', 5'-phosphodiester bond.

The phosphodiester bond (pale peach shading) joins the 3' carbon atom of one sugar to the 5' carbon atom of the next sugar in the sugar-phosphate backbone of a nucleic acid.

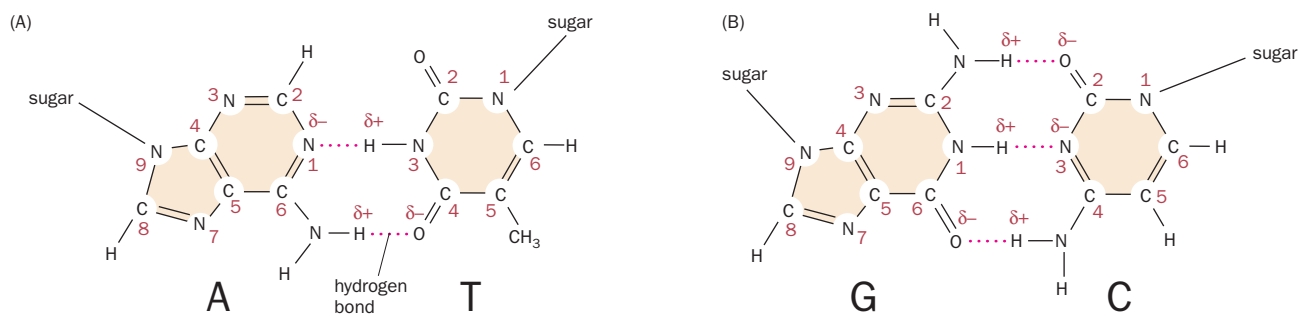


Figure 1.6 AT and GC base pairs. (A) AT base pairs have two connecting hydrogen bonds (dotted red lines); (B) GC base pairs have three. Fractional positive charges and fractional negative charges are shown by $\delta+$ and $\delta-$, respectively.

Because the phosphodiester bonds link carbon atoms number 3' and number 5' of successive sugar residues, the two ends of a linear DNA strand are different. The **5' end** has a terminal sugar residue in which carbon atom number 5' is not linked to another sugar residue. The **3' end** has a terminal sugar residue whose 3' carbon is not involved in phosphodiester bonding. The two strands of a DNA duplex are described as being anti-parallel to each other because the 5'→3' direction of one DNA strand is the opposite to that of its partner, according to Watson–Crick base-pairing rules (**Figure 1.8**).

Genetic information is encoded by the linear sequence of bases in the DNA strands. The two strands of a DNA duplex have complementary sequences, so the sequence of bases of one DNA strand can therefore readily be inferred from that of the other strand. It is usual to describe DNA by writing the sequence of bases of one strand only, in the 5'→3' direction, which is the direction of synthesis of new DNA or RNA from a DNA template. When describing the sequence of a DNA region encompassing two neighboring bases (a dinucleotide) on one DNA strand, it is usual to insert a 'p' to denote a connecting phosphodiester bond. So, a CG base pair means a C on one DNA strand is hydrogen-bonded to a G on the complementary strand, but CpG represents a deoxycytidine covalently linked to a neighboring deoxyguanosine on the same DNA strand (see **Figure 1.8**).

Unlike DNA, RNA is normally single-stranded except for certain viruses that have double-stranded RNA genomes. However, to perform certain cell functions two RNA molecules may need to associate transiently to form base pairs, and intermolecular hydrogen bonding also permits the formation of transient RNA–DNA duplexes (see **Box 1.1**).

In addition, hydrogen bonding can occur between bases within a single-stranded RNA (or DNA) molecule to produce structurally and functionally important stretches of double-stranded sequence. Hairpin structures may be formed with stems that are stabilized by hydrogen bonding between bases (**Figure 1.9A**). Intrachain base pairing causes certain RNA molecules to have complex structures (**Figure 1.9B**).

In double-stranded RNA, A pairs with U instead of T. Although G usually pairs with C, sometimes G–U base pairs are formed (see the example in **Figure 1.9B**). Although not particularly stable, G–U base pairing does not significantly distort the RNA–RNA helix.

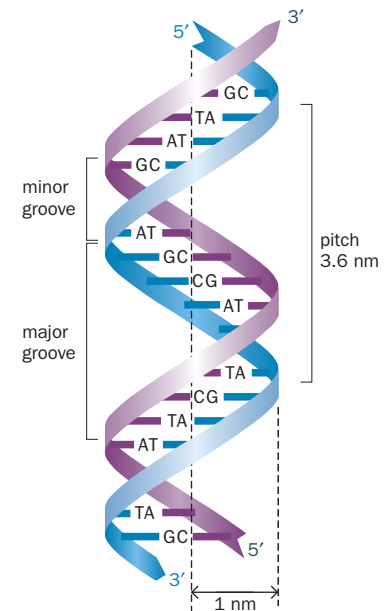


Figure 1.7 Features of the DNA double helix. The two DNA strands wind round each other, producing a minor groove and a major groove in the double helix. The double helix has a pitch of 3.6 nm and a radius of 1 nm per turn.

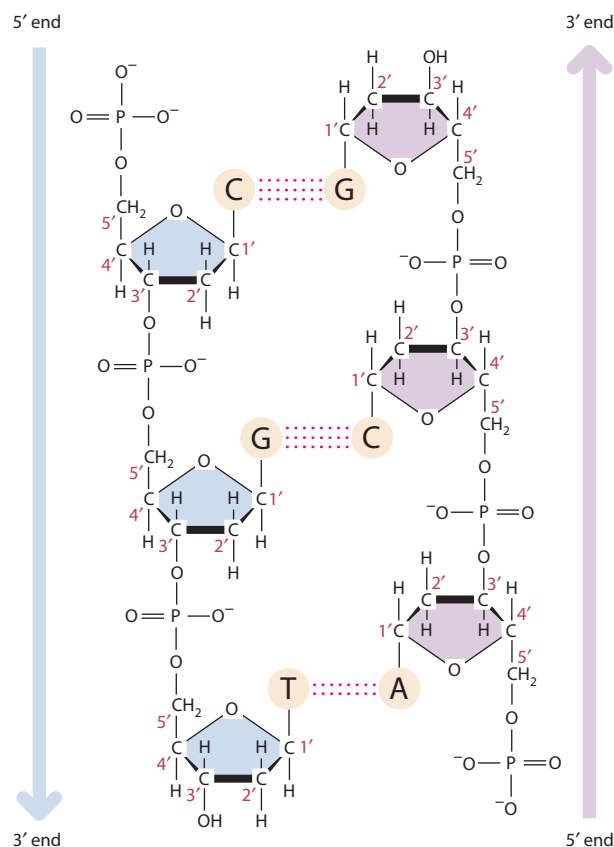


Figure 1.8 Anti-parallel nature of the DNA double helix. The two anti-parallel DNA strands run in opposite directions in linking 3' to 5' carbon atoms in the sugar residues. This double-stranded trinucleotide has the sequence 5' pCpGpT–OH 3'/5' pApCpG–OH 3', where p stands for a phosphate group and –OH 3' represents the 3' terminal hydroxyl group. This is conventionally abbreviated to give the 5'→3' sequence of nucleotides on only one strand, either as 5'–CGT–3' (blue strand) or as 5'–ACG–3' (purple strand).

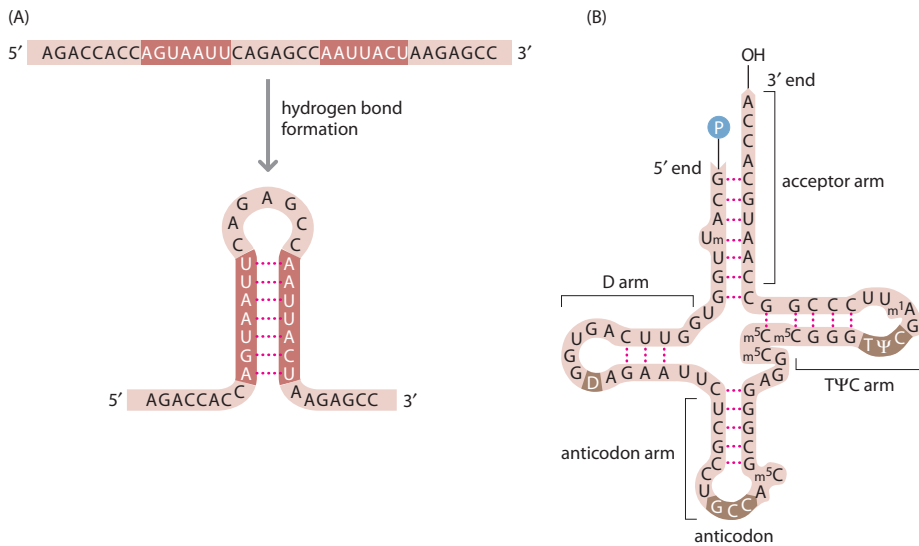


Figure 1.9 Base pairing within single-stranded nucleic acids. (A) Hairpin formation by intramolecular hydrogen bonding. Hydrogen bonds between the sequences highlighted by dark pink shading within this single-stranded nucleic acid (shown here as RNA) can stabilize folding back to form a hairpin with a double-stranded stem. (B) Extensive intramolecular base pairing in transfer RNA. The tRNA^{Gly} shown here as an example illustrates the classical cloverleaf tRNA structure. There are three hairpins (the D arm, anticodon arm, and the TΨC arm) plus a stretch of base pairing between 5' and 3' terminal sequences (called the acceptor arm because the 3' end is used to attach an amino acid). Note that tRNAs always have the same number of base pairs in the stems of the different arms of their cloverleaf structure and that the anticodon at the center of the middle loop identifies the tRNA according to the amino acid it will bear. The minor nucleotides depicted are: D, 5,6-dihydrouridine; Ψ, pseudouridine (5-ribosyluracil); m⁵C, 5-methylcytidine; m¹A, 1-methyladenosine; Um, 2'-O-methyluridine.

Replication is semi-conservative and semi-discontinuous

For new DNA synthesis (**replication**) to begin, the two DNA strands of a helix need to be unwound by the enzyme helicase. The two unwound DNA strands then each serve as a template for DNA polymerase to make complementary DNA strands, using the four deoxynucleoside triphosphates (dATP, dCTP, dGTP, and dTTP). Two daughter DNA duplexes are formed, each identical to the parent molecule (**Figure 1.10**). Each daughter DNA duplex contains one strand from the parent molecule and one newly synthesized DNA strand, so the replication process is **semi-conservative**.

DNA replication is initiated at specific points, called **origins of replication**, generating Y-shaped replication forks, where the parental DNA duplex is opened up. The anti-parallel parental DNA strands serve as templates for the synthesis of complementary daughter strands that run in opposite directions.

The overall direction of chain growth is 5'→3' for one daughter strand, the **leading strand**, but 3'→5' for the other daughter strand, the **lagging strand** (**Figure 1.11**). The reactions catalyzed by DNA polymerase involve the addition of a deoxynucleoside monophosphate (dNMP) residue to the free 3' hydroxyl group of the growing DNA strand. However, only the leading strand always has a free 3' hydroxyl group that allows continuous elongation in the same direction in which the replication fork moves.

The direction of synthesis of the lagging strand is opposite to that in which the replication fork moves. As a result, strand synthesis needs to be accomplished in a progressive series of steps, making DNA segments that are typically 100–1000 nucleotides long (Okazaki fragments). Successively synthesized fragments are eventually joined covalently by the enzyme DNA ligase to ensure the creation of two complete daughter DNA duplexes. Only the leading strand is synthesized continuously, so DNA synthesis is therefore **semi-discontinuous**.

DNA polymerases sometimes work in DNA repair and recombination

The machinery for DNA replication relies on a variety of proteins (**Box 1.2**) and RNA primers, and has been highly conserved during evolution. However, the complexity of the process is greater in mammalian cells, in terms of the numbers of different DNA polymerases (**Table 1.3**), and of their constituent proteins and subunits.

Most DNA polymerases in mammalian cells use an individual DNA strand as a template for synthesizing a complementary DNA strand and so are DNA-directed DNA polymerases. Unlike RNA polymerases, DNA polymerases normally require the 3'-hydroxyl end of a base-paired primer strand as a substrate. Therefore, an RNA primer, synthesized by a primase, is needed to provide a free 3' OH group for the DNA polymerase to start synthesizing DNA.

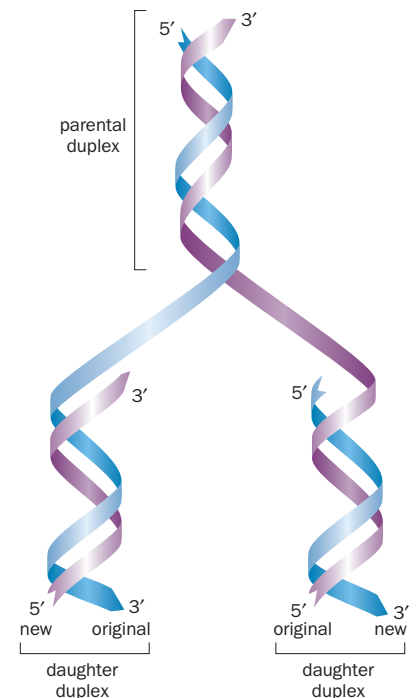


Figure 1.10 Semi-conservative DNA replication. The parental DNA duplex consists of two complementary, anti-parallel DNA strands that unwind to serve as templates for the synthesis of new complementary DNA strands. Each completed daughter DNA duplex contains one of the two parental DNA strands plus one newly synthesized DNA strand, and is structurally identical to the original parental DNA duplex.

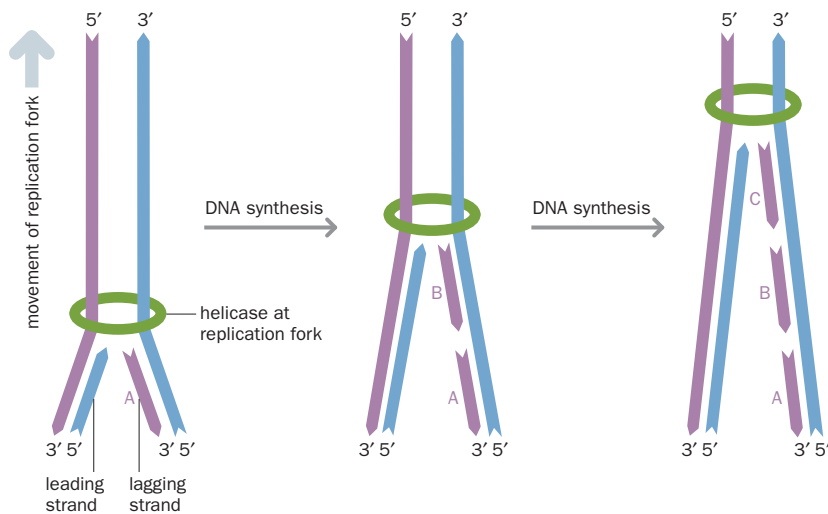


Figure 1.11 Semi-discontinuous DNA replication. The enzyme helicase opens up a replication fork, where synthesis of new daughter DNA strands can begin. The overall direction of movement of the replication fork matches that of the continuous 5'→3' synthesis of the leading daughter DNA strand. Replication is semi-discontinuous because the lagging strand, which is synthesized in the opposite direction, is built up in pieces (Okazaki fragments, shown here as fragments A, B, and C), that will later be stitched together by a DNA ligase.

There are close to 20 different types of DNA polymerase in mammalian cells. Most use DNA as a template to synthesize DNA and they have been grouped into four families—A, B, X, and Y—on the basis of sequence comparisons (see Table 1.3).

Members of family B are classical (high-fidelity) DNA polymerases and include the enzymes devoted to replicating nuclear DNA. They mostly have an associated 3'-5' exonuclease activity that is important in **proofreading**: if the wrong base is inserted at the 3' OH group of the growing DNA chain the 3'-5' exonuclease snips it out. This results in high-fidelity replication, because base misincorporation errors are extremely infrequent. DNA polymerase α is a complex of a polymerase and a primase and is devoted to initiating DNA synthesis and initiating Okazaki fragments. DNA polymerases δ and ϵ carry out most of the DNA synthesis and are strand-specific (see Table 1.3).

Many DNA polymerases work in DNA repair or recombination. They include classical high-fidelity DNA polymerases that are also involved in replication (DNA polymerases δ and ϵ) and others that are dedicated to DNA repair or recombination. Some of the latter are high-fidelity polymerases but many of them are comparatively prone to base misincorporation, notably members of family X and especially family Y members. For example, DNA polymerase ι (iota) can have an error rate 20,000 or more times that of DNA polymerase ϵ .

The high error rate in some DNA polymerases is tolerated because they work in DNA repair processes and so are used to synthesize only small stretches of DNA. In other cases, high error rates are advantageous. For example, low-fidelity

BOX 1.2 MAJOR CLASSES OF PROTEINS INVOLVED IN DNA REPLICATION

- **Topoisomerases**—start the process of DNA unwinding by breaking a single DNA strand, releasing the tension holding the helix in its coiled and supercoiled form.
- **Helicases**—unwind the double helix at the replication fork, once supercoiling has been eliminated by a topoisomerase.
- **Single-stranded binding proteins**—maintain the stability of the replication fork. Single-stranded DNA is very vulnerable to enzymatic attack; the bound proteins protect it from being degraded.
- **Primases**—enzymes that attach a small complementary RNA sequence (a **primer**) to single-stranded DNA at the replication fork. The RNA primer provides the 3' OH needed by DNA polymerase to begin synthesis (unlike RNA polymerases, DNA polymerases cannot initiate new strand synthesis from a bare single-stranded template but require an initiating molecule with a free 3' OH group onto which deoxynucleoside triphosphates can be attached to build a complementary strand).
- **DNA polymerases**—for synthesizing new DNA strands. New cellular DNA synthesis normally depends on an existing DNA strand template that is read by a DNA-directed DNA polymerase. This complex aggregate of protein subunits often also provides DNA proofreading and DNA repair functions (see Table 1.3). This means that any wrongly incorporated bases can be identified, removed, and repaired. DNA can also be synthesized from an RNA template, using an RNA-directed DNA polymerase (a reverse transcriptase). The ends of linear chromosomes are copied using a reverse transcriptase (telomerase).
- **DNA ligases**—needed to seal nicks that remain in newly synthesized DNA after the RNA primers have been removed and the small gaps filled by DNA polymerase. The DNA ligases catalyze the formation of a phosphodiester bond between unattached but adjacent 3' hydroxyl and 5' phosphate groups.

TABLE 1.3 MAMMALIAN DNA POLYMERASES

DNA-DIRECTED DNA POLYMERASES			
Polymerase	Family	Standard DNA replication	Additional or alternative roles in DNA repair, recombination, etc.
α (alpha)	B	initiates synthesis at replication origins and initiates synthesis of Okazaki fragments on lagging strand	
β (beta)	X		base excision repair ^b
γ (gamma)	A	mitochondrial DNA synthesis	mitochondrial DNA repair
δ (delta)	B	main polymerase that synthesizes lagging strand	multiple roles in DNA repair
ϵ (epsilon)	B	synthesizes leading strand	multiple roles in DNA repair
ζ (zeta)	B		translesion synthesis ^c
η (eta)	Y		translesion synthesis ^c
θ (theta)	A		possible role in interstrand crosslink repair ^d ; base excision repair ^b ; translesion synthesis ^c ; somatic hypermutation ^g
ι (iota)	Y		translesion synthesis ^c ; possible roles in base excision repair ^b and mismatch repair ^e
κ (kappa)	Y		translesion synthesis ^c ; nucleotide excision repair ^f
λ (lambda)	X		double-strand break repair; VDJ recombination ^g ; base excision repair ^b
μ (mu)	X		
ν (nu)	A		possible role in interstrand crosslink repair ^d
Rev1	Y		translesion synthesis ^c
TdT ^a	X		VDJ recombination ^g
RNA-DIRECTED DNA POLYMERASES (REVERSE TRANSCRIPTASES)			
Interspersed repeat reverse transcriptases (LINE-1 or endogenous retrovirus elements)		occasionally converts mRNA and other RNA into cDNA, which can integrate elsewhere into the genome	
Telomerase reverse transcriptase (Tert)		replicates DNA at the ends of linear chromosomes	

^aTerminal deoxynucleotide transferase. ^bBase excision repair identifies and removes inappropriate bases or inappropriately modified bases.

^cTranslesion synthesis involves the replication of DNA past damaged DNA (lesions) on the template strand. ^dInterstrand crosslink repair is the repair of highly cytotoxic lesions where covalent DNA bonds have been formed between the DNA strands. ^eMismatch repair is a form of DNA repair that corrects mistakes arising when noncomplementary nucleotides form a base pair. ^fNucleotide excision repair is used to fix helix-distorting lesions.

^gSomatic hypermutation and VDJ recombination are mechanisms used in B cells to diversify immunoglobulin sequences.

DNA polymerases can continue to synthesize new DNA strands opposite a lesion in the template DNA (translesion synthesis) and they can contribute to the sequence diversity of immunoglobulins (e.g. by introducing many base changes in coding sequences) and so assist in the recognition of numerous foreign antigens by the immune system.

Many viruses have RNA genomes

DNA is the hereditary material in all present-day cells and we generally think of the **genome** as the collective term for the different hereditary DNA molecules of an organism or cell. However, many viruses have an RNA genome. These RNA molecules can undergo self-replication, although the 2' OH group on their ribose residues makes the sugar-phosphate bonds rather unstable chemically. By contrast, in DNA, the deoxyribose residues carry only hydrogen atoms at the 2' position, making DNA a more stable carrier of genetic information.

TABLE 1.4 DIFFERENT CLASSES OF GENOME

	Single linear	Multiple linear	Single circular	Multiple circular	Mixed (linear + circular)
DNA GENOMES					
Single-stranded (ss)DNA	some viruses	segmented ssDNA viruses	some viruses	–	–
Double-stranded (ds)DNA	some viruses; a very few bacteria, e.g. <i>Borrelia</i>	segmented dsDNA viruses; eukaryotic nuclei	mitochondria; chloroplasts; many bacteria and Archaea	multipartite viruses; some bacteria	a very few bacteria, e.g. <i>Agrobacterium tumefaciens</i>
RNA GENOMES					
Single-stranded (ss)RNA	some viruses	segmented ssRNA viruses	a very few viruses	–	–
Double-stranded (ds)RNA	a few viruses	segmented dsRNA viruses	–	–	–

See Figure 1.12 for examples of viral genomes and for explanations of segmented and multipartite viruses.

Viruses have developed many different strategies to infect and subvert cells, and their genomes show extraordinary diversity when compared with cellular genomes (**Table 1.4** and **Figure 1.12**). Because RNA replication has a much higher error rate than DNA replication, viral RNA genomes have a higher mutational load than DNA genomes. Although viral RNA genomes are generally quite small, the elevated mutation rate permits more rapid adaptation to changing environmental conditions. RNA viruses usually replicate in the cytoplasm; DNA viruses generally replicate in the nucleus.

Retroviruses are unusual RNA viruses both because they replicate in the nucleus and also because their RNA replicates via a DNA intermediate. The single-stranded RNA genome is converted into a single-stranded cDNA using a viral reverse transcriptase. The single-stranded viral cDNA is then converted into double-stranded DNA, by using a DNA polymerase from the host cell. Other viral proteins then help insert this double-stranded DNA into the host cell's chromosomal DNA. It can remain there for long periods or be used to synthesize new viral RNA genomes that are packaged as new virus particles.

1.3 RNA TRANSCRIPTION AND GENE EXPRESSION

As well as having global roles in storing and transmitting genetic information and supporting chromosome function, DNA can have cell-specific functions because it contains sequences that can be used to make RNA and polypeptides in ways that differ from cell to cell. **Genes** are discrete DNA segments that are spaced at irregular intervals along the DNA sequence and serve as templates for making complementary RNA sequences (transcription). The initial primary RNA transcript must then undergo a series of maturation steps that ultimately result in a mature, functional noncoding RNA or a messenger RNA that will in turn serve as a template to make a polypeptide. Some of the gene products are needed by essentially all cells for a variety of vital cell processes (such as DNA replication or protein synthesis). However, other RNA and protein products are made in some cell types but not others and may even be specific for individual cells in some exceptional cases, as in individual B and T lymphocytes.

The DNA compositions of the different cell types in a multicellular organism are essentially identical. The variation between cells happens because of differences in gene expression, primarily at the level of transcription: different genes are transcribed in different cells according to the needs of the cells. Some genes, known as housekeeping genes, need to be expressed in essentially all cells, but other genes show tissue-specific gene expression or they may be expressed at specific times (e.g. at specific stages of development or of the cell cycle).

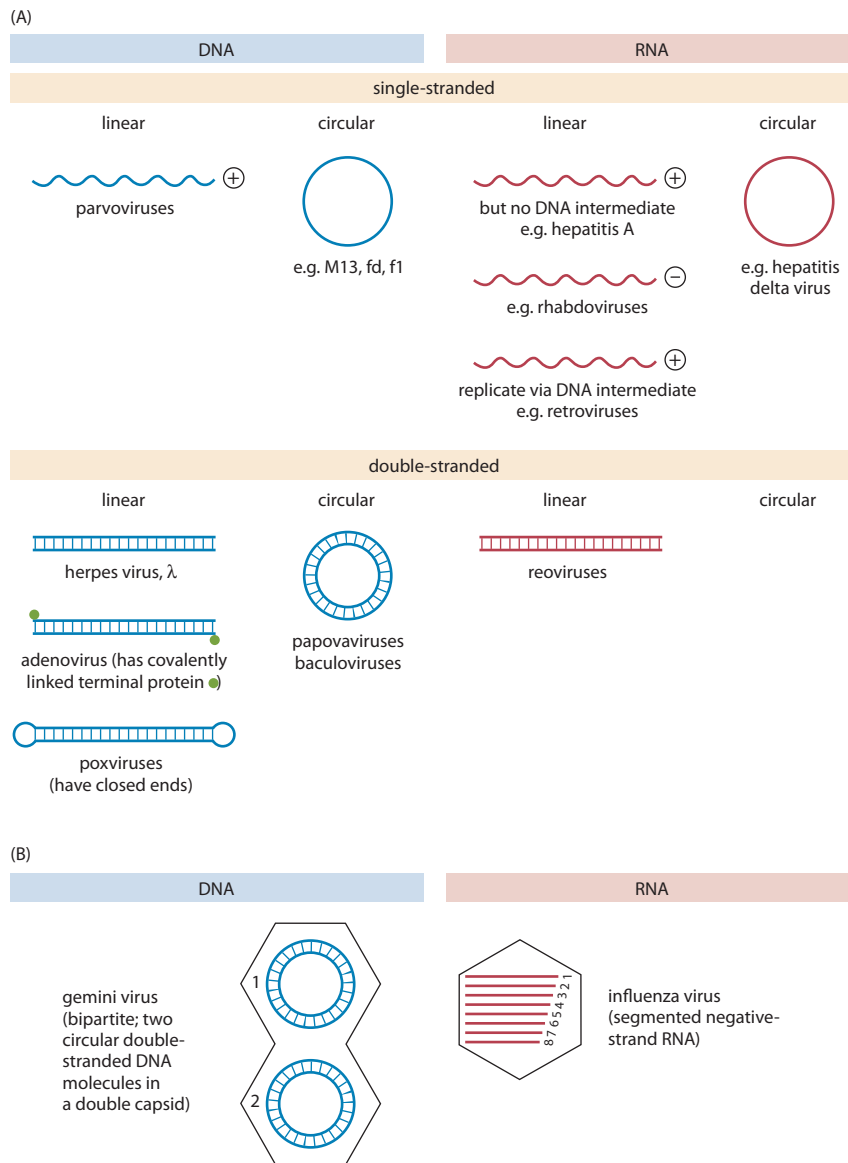


Figure 1.12 The extraordinary variety of viral genomes. (A) Strandedness and topology. In single-stranded viral genomes, the RNA used to make protein products may have the same sense as the genome, which is therefore a *positive-strand genome* (+), or be the opposite sense (antisense) of the genome, a *negative-strand genome* (-). Some single-stranded (+) RNA viruses (e.g. retroviruses) go through a DNA intermediate, and some double-stranded DNA viruses (e.g. hepatitis B) go through a replicative RNA form. (B) Segmented and multipartite genomes. *Segmented genomes* are ones in which the genome is divided into multiple different nucleic acid molecules, each specifying a messenger RNA devoted to making a single polypeptide. For example, the genome of an influenza virus is partitioned into eight different negative single-strand RNA molecules. In some segmented genomes, each of the different molecules is packaged in a separate virus particle (capsid). Such genomes are described as *multipartite genomes*, as illustrated here by the bipartite genome of the gemini virus.

Normally, only one of the two DNA strands in a duplex serves as a template for RNA synthesis. During transcription, double-stranded DNA is bound by RNA polymerase. The DNA is then unwound, enabling the DNA strand that will act as a template for RNA synthesis (the template strand) to form a transient double-stranded RNA–DNA hybrid with the growing RNA strand.

The RNA transcript is complementary to the template strand of the DNA and has the same 5'→3' direction and base sequence (except that U replaces T) as the opposite, nontemplate DNA strand. The nontemplate strand is often called the **sense strand**, and the template strand is often called the **antisense strand** (Figure 1.13).

In documenting gene sequences, it is customary to show only the DNA sequence of the sense strand. The orientation of sequences relative to a gene normally refers to the sense strand. For example, the 5' end of a gene refers to

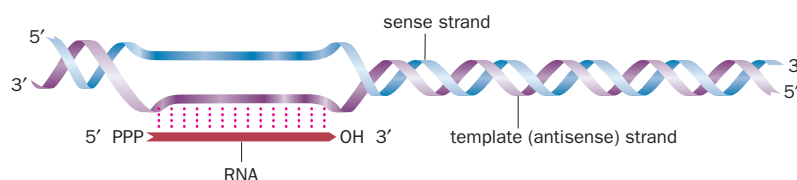


Figure 1.13 Transcribed RNA is complementary to one strand of DNA.

The nucleotide sequence of transcribed RNA is normally identical to that of the sense strand, except that U replaces T, and is complementary to that of the template strand. The nucleotide at the extreme 5' end of a primary RNA transcript carries a 5' triphosphate group that may later undergo modification; the 3' end has a free hydroxyl group.

sequences at the 5' end of the sense strand, and upstream or downstream sequences flank the gene at its 5' or 3' ends, respectively, with reference to the *sense* strand. For transcription to proceed efficiently, various proteins (transcription factors) must bind to particular DNA sequence elements (collectively called a **promoter**) that are often located close to and upstream of a gene. The bound transcription factors serve to position and guide the RNA polymerase.

RNA polymerases synthesize RNA from four nucleotide precursors: ATP, CTP, GTP, and UTP. Elongation involves the addition of the appropriate ribonucleoside monophosphate residue (AMP, CMP, GMP, or UMP) to the free 3' hydroxyl group at the 3' end of the growing RNA strand. These nucleotides are derived by splitting a pyrophosphate residue (PP_i) from their appropriate ribonucleoside triphosphate (rNTP) precursors. Only the initiator nucleotide at the extreme 5' end of a primary transcript carries a 5' triphosphate group.

Most genes are expressed to make polypeptides

Most eukaryotic genes are expressed to produce polypeptides using RNA polymerase II, one of three RNA polymerases (Table 1.5). All three RNA polymerases cannot initiate transcription by themselves: they require regulatory factors. A crucial regulatory element is the **promoter**, a collection of closely spaced short DNA sequence elements in the immediate vicinity of a gene. Promoters are recognized and bound by transcription factors that then guide and activate the polymerase. Transcription factors are said to be **trans-acting**, because they are produced by remote genes and need to migrate to their sites of action. In contrast, promoter sequences are **cis-acting** because they are located on the same DNA molecule as the genes they regulate.

Promoters recognized by RNA polymerase II often include the following elements:

- The **TATA box**. Often TATAAA, or a variant sequence, this element is usually found about 25 base pairs (bp) upstream from the transcriptional start site (designated by –25; Figure 1.14A). It usually occurs in genes that are actively transcribed by RNA polymerase II only at a particular stage in the cell cycle (e.g. histone genes) or in specific cell types (e.g. the β -globin gene). A mutation in the TATA box does not prevent the initiation of transcription but causes transcription to begin at an incorrect location.
- The **GC box**. Usually a variant of the sequence GGGCGG, the GC box occurs in a variety of genes, many of which lack a TATA box. This is the case for the 'housekeeping' genes that perform the same function in all cells (such as those encoding DNA and RNA polymerases, histones, or ribosomal proteins). Although the GC box sequence is sequentially asymmetrical, it seems to function in either orientation (Figure 1.14B).
- The **CAAT box**. Often located at position –80 it is usually the strongest determinant of promoter efficiency. Like the GC box, it functions in either orientation.

TABLE 1.5 THE THREE CLASSES OF EUKARYOTIC RNA POLYMERASE

RNA polymerase	RNA synthesized	Notes
I	28S rRNA ^a , 18S rRNA ^a , 5.8S rRNA ^a	localized in the nucleolus; RNA polymerase I produces a single primary transcript (45S rRNA) that is cleaved to give the three rRNA classes listed here
II	mRNA ^b , miRNA ^c , most snRNAs ^d and snoRNAs ^e	RNA polymerase II transcripts are unique in being subject to capping and polyadenylation
III	5S rRNA ^a , tRNA ^f , U6 snRNA ^g , 7SL RNA ^h , various other small noncoding RNAs	the promoter for some genes transcribed by RNA polymerase III (e.g. 5S rRNA, tRNA, 7SL RNA) is internal to the gene; for others, it is located upstream of the gene (see Figure 1.15)

^aRibosomal RNA. ^bMessenger RNA. ^cMicroRNA. ^dSmall nuclear RNAs. ^eSmall nucleolar RNAs. ^fTransfer RNA. ^gU6 snRNA is a component of the spliceosome, an RNA–protein complex that removes unwanted noncoding sequences from newly formed RNA transcripts. ^h7SL RNA forms part of the signal recognition particle, which has an important role in the transport of newly synthesized proteins.

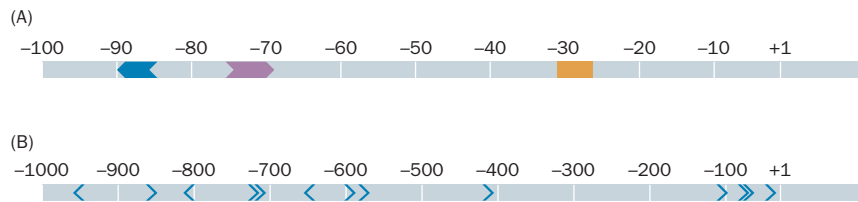


Figure 1.14 Promoters for two eukaryotic genes encoding polypeptides. Polypeptide-encoding genes are transcribed by RNA polymerase II. The promoters are defined by short sequence elements located in regions just upstream of the transcription start site (+1). (A) The β -globin gene promoter includes a TATA box (orange), a CAAT box (purple), and a GC box (blue). (B) The glucocorticoid receptor gene is unusual in possessing 13 upstream GC boxes: 10 in the normal orientation, and 3 in the reverse orientation (alternative orientations for GC box elements are indicated by chevron directions).

For a gene to be transcribed by RNA polymerase II the DNA must first be bound by general transcription factors, to form a preinitiation complex. General transcription factors required by RNA polymerase II include TFIIA, TFIIB, TFIID, TFIIIE, TFIIF, and TFIIF. These transcription factors may themselves comprise several components. For example, TFIID consists of the TATA-box-binding protein (TBP; also found in association with RNA polymerases I and III) plus various TBP-associated factors (TAF proteins). The complex that is required to initiate transcription by an RNA polymerase is known as the **basal transcription apparatus** and consists of the polymerase plus all of its associated general transcription factors.

In addition to the general transcription factors required by RNA polymerase II, specific recognition elements are recognized by tissue-restricted transcription factors. For example, an **enhancer** is a cluster of *cis*-acting short sequence elements that can enhance the transcriptional activity of a specific eukaryotic gene. Unlike a promoter, which has a relatively constant position with regard to the transcriptional initiation site, enhancers are located at variable (often considerable) distances from their transcriptional start sites. Furthermore, their function is independent of their orientation. Enhancers do, however, also bind gene regulatory proteins. The DNA between the promoter and enhancer sites loops out, which brings the two different DNA sequences together and allows the proteins bound to the enhancer to interact with the transcription factors bound to the promoter, or with the RNA polymerase.

A silencer has similar properties to an enhancer but it inhibits, rather than stimulates, the transcriptional activity of a specific gene.

Different sets of RNA genes are transcribed by the three eukaryotic RNA polymerases

Genes that encode polypeptides are always transcribed by RNA polymerase II. However, RNA genes (genes that make noncoding RNA) may be transcribed by polymerases I, II, or III, depending on the type of RNA (see Table 1.5). RNA polymerase I is unusual because it is dedicated to transcribing RNA from a single transcription unit, generating a large transcript that is then processed to yield three types of ribosomal RNA (see below).

RNA polymerase II synthesizes various types of small noncoding RNA in addition to mRNA. They include many types of small nuclear RNA (snRNA) and of small nucleolar RNA (snoRNA) that are involved in different RNA processing events. In addition, it synthesizes many microRNAs (miRNAs) that can show tissue-specific expression and typically regulate the expression of distinctive sets of target genes.

RNA polymerase III transcribes a variety of small noncoding RNAs that are typically expressed in almost all cells, including the different transfer RNA species, 5S ribosomal RNA (rRNA), and some snRNAs. The genes for transfer RNAs (tRNAs) and 5S rRNA are unusual in that the promoters lie within, rather than upstream of, the transcribed sequence (Figure 1.15).

Internal promoters are possible because the job of a promoter is simply to attract transcription factors that will guide the RNA polymerase to the correct transcriptional start site. By the time the polymerase is in place and ready to initiate transcription, any transcription factors previously bound to downstream promoter elements will have been removed from the template strand. As an example, transcription of a tRNA gene begins with the following sequence:

- TFIIC (transcription factor for polymerase III) binds to the A and B boxes of the internal promoter of a tRNA gene (see Figure 1.15).

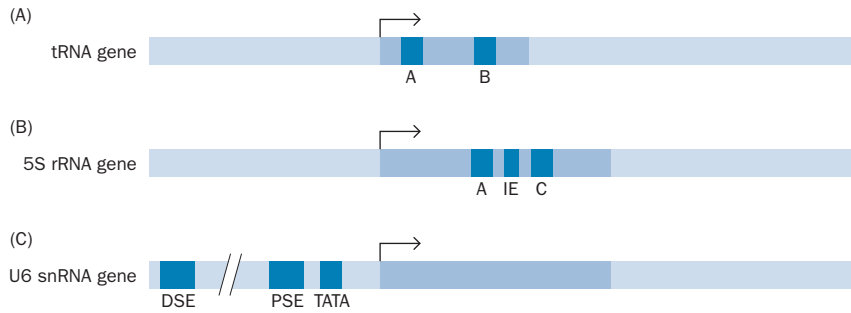


Figure 1.15 Promoter elements in three genes transcribed by RNA polymerase III.

(A) tRNA genes have an internal promoter consisting of an A box (located within the D arm of the tRNA; see Figure 1.9B) and a B box that is usually found in the TψC arm. (B) The promoter of the *Xenopus* 5S rRNA gene has three components: an A box (+50 to +60), an intermediate element (IE; +67 to +72), and the C box (+80 to +90). (C) The human U6 snRNA gene has an external promoter consisting of three components. A distal sequence element (DSE; -240 to -215) enhances transcription and works alongside a core promoter composed of a proximal sequence element (PSE; -65 to -48) and a TATA box (-32 to -25). Arrows mark the +1 position.

- Bound TFIIC guides the binding of another transcription factor, TFIIB, to a position upstream of the transcriptional start site; TFIIC is no longer required and any bound TFIIC is removed from the internal promoter.
- TFIIB guides RNA polymerase III to bind to the transcriptional start site.

1.4 RNA PROCESSING

The RNA transcript of most eukaryotic genes undergoes a series of processing reactions to make a mature mRNA or noncoding RNA.

RNA splicing removes unwanted sequences from the primary transcript

For most vertebrate genes—almost all protein-coding genes and some RNA genes—only a small portion of the gene sequence is eventually decoded to give the final product. In these cases the genetic instructions for making an mRNA or mature noncoding RNA occur in **exon** segments that are separated by intervening **intron** sequences that do not contribute genetic information to the final product.

Transcription of a gene initially produces a primary transcript RNA that is complementary to the entire length of the gene, including both exons and introns. This primary transcript then undergoes **RNA splicing**, which is a series of reactions whereby the intronic RNA segments are removed and discarded while the remaining exonic RNA segments are joined end-to-end, to give a shorter RNA product (Figure 1.16).

RNA splicing requires recognition of the nucleotide sequences at the boundaries of transcribed exons and introns (splice junctions). The dinucleotides at the ends of introns are highly conserved: the vast majority of introns start with a GT (becoming GU in intronic RNA) and end with an AG (the **GT-AG rule**).

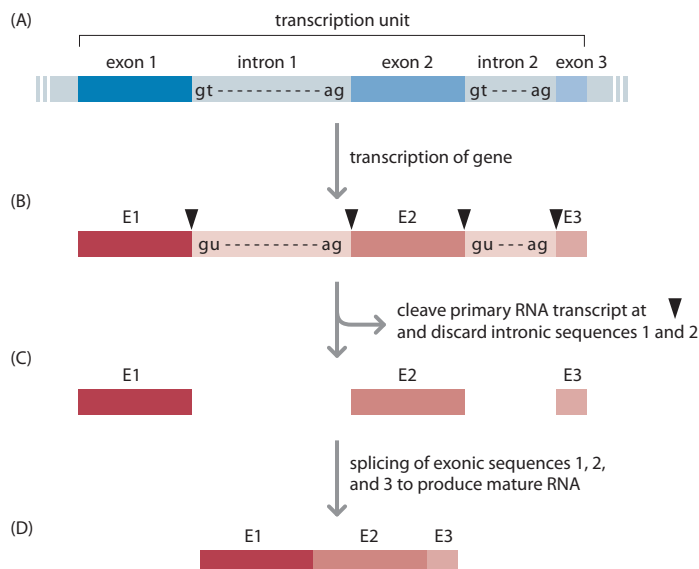
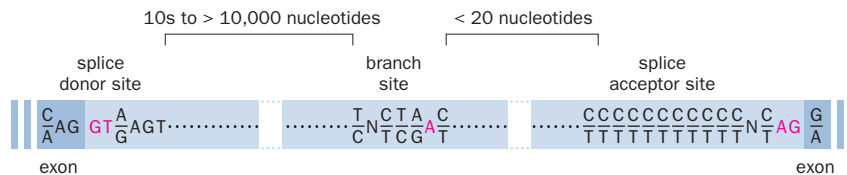


Figure 1.16 The process of RNA splicing.

(A) In this example, the gene contains three exons and two introns. (B) The primary RNA transcript is a continuous RNA copy of the gene and contains sequences transcribed from exons (E1, E2, and E3) and introns. (C) The primary transcript is cleaved at regions corresponding to exon-intron boundaries (splice junctions). The RNA copies of the introns are snipped out and discarded. (D) The RNA copies of the exons are retained and then fused together (spliced) in the same linear order as in the genomic DNA sequence.



Although the conserved GT and AG dinucleotides are crucial for splicing, they are not sufficient to mark the limits of an intron. The nucleotide sequences that are immediately adjacent to them are also quite highly conserved, constituting splice junction consensus sequences (Figure 1.17). A third conserved intronic sequence that is also important in splicing is known as the branch site and is typically located no more than 40 nucleotides upstream of the intron's 5' terminal AG (see Figure 1.17). Other exonic and intronic sequences can promote splicing (splice enhancer sequences) or inhibit it (splice silencer sequences), and mutations in these sequences can cause disease.

The essential steps in splicing are as follows:

- Nucleophilic attack of the intron's 5' terminal G nucleotide by the invariant A of the branch site consensus sequence, to form a lariat-shaped structure.
- Cleavage of the exon/intron junction at the splice donor site.
- Nucleophilic attack by the 3' end of the upstream exon of the splice acceptor site, leading to cleavage and release of the intronic RNA in the form of a lariat, and the splicing together of the two exonic RNA segments (Figure 1.18).

For genes residing in eukaryotic nuclei, RNA splicing is mediated by a large RNA-protein complex, called the spliceosome. Spliceosomes have five types of snRNA (small nuclear RNA) and more than 50 proteins. The snRNA molecules associate with proteins to form small nuclear ribonucleoprotein (snRNP, or snurp) particles. The specificity of the splicing reaction is established by RNA-RNA base pairing between the RNA transcript to be spliced and snRNA molecules within the spliceosome. There are two types of spliceosome:

- The *major (GU-AG) spliceosome* processes transcripts corresponding to classical GT-AG introns. It contains five types of snRNA. U1 and U2 snRNAs recognize and bind the splice donor and branch sites, respectively. U4, U5, and U6 snRNAs subsequently bind to cause looping out of the intronic RNA (Figure 1.19).
- The *minor (AU-AC) spliceosome* processes transcripts corresponding to rare AU-AC introns. It also has five snRNAs but uses U11 and U12 snRNA instead of U1 and U2 and has variants of U4 and U6 snRNA.

Once a splice donor site is recognized by the spliceosome, it scans the RNA sequence until it meets the next splice acceptor site (signaled as a target by the upstream presence of the branch site consensus sequence).

Specialized nucleotides are added to the ends of most RNA polymerase II transcripts

In addition to RNA splicing, the ends of RNA polymerase II transcripts undergo modifications: the 5' end is capped by adding a variant guanine by using an unusual phosphodiester bond, and a long sequence of adenines is added to the 3' end. As well as protecting the ends from cellular exonucleases, these modifications may assist the correct functioning of the RNA transcripts.

Figure 1.18 The mechanism of RNA splicing. (A) The unprocessed primary RNA transcript with intronic RNA separating sequences E1 and E2 that correspond to exons in DNA. The splicing mechanism involves a nucleophilic attack on the G of the 5' GU dinucleotide. This is carried out by the 2' OH group on the conserved A of the branch site and results in the formation of a lariat structure (B), and cleavage of the splice donor site. The 3' OH at the 3' end of the E1 sequence performs a nucleophilic attack on the splice acceptor site, causing release of the intronic RNA (as a lariat-shaped structure) and (C) fusion (splicing) of E1 and E2.

Figure 1.17 Three consensus DNA sequences in introns of complex eukaryotes. Most introns in eukaryotic genes contain conserved sequences that correspond to three functionally important regions. Two of the regions, the splice donor site and the splice acceptor site, span the 5' and 3' boundaries of the intron. The branch site is an additional important region that typically occurs less than 20 nucleotides upstream of the splice acceptor site. The nucleotides shown in red in these three consensus sequences are almost invariant. The other nucleotides detailed in both the intron and the exons are those most commonly found at each position. In some instances, two nucleotides may be equally common, as in the case of C and T near the 3' end of the intron. Where N appears, any of the four nucleotides may occur.

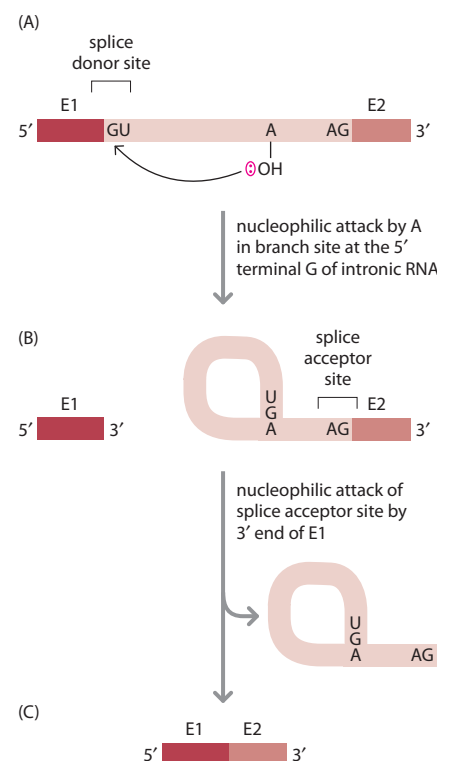
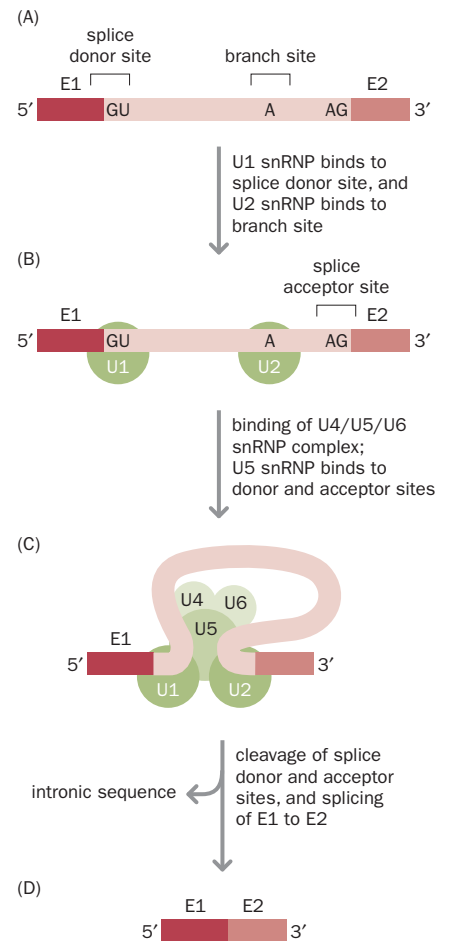


Figure 1.19 Role of small nuclear ribonucleoprotein (snRNPs) in RNA splicing.

(A) The unprocessed primary RNA transcript as in Figure 1.18. (B) Within the spliceosome, part of the U1 snRNA is complementary in sequence to the splice donor site consensus sequence. As a result, the U1 snRNA-protein complex (U1 snRNP) binds to the splice junction by RNA–RNA base pairing. The U2 snRNP complex similarly binds to the branch site by RNA–RNA base pairing. Interaction between the splice donor and splice acceptor sites is stabilized by (C) the binding of a multi-snRNP particle that contains the U4, U5, and U6 snRNAs. The U5 snRNP binds simultaneously to both the splice donor and splice acceptor sites. Their cleavage releases the intronic sequence and allows (D) E1 and E2 to be spliced together.



5' capping

Shortly after the initiation of synthesis of primary RNA transcripts that will become mRNA, a methylated nucleoside (7-methylguanosine, m⁷G) is linked by a 5'–5' phosphodiester bond to the first 5' nucleotide. This is described as the **capping** of the 5' end of the transcript (Figure 1.20); the caps of snRNA gene transcripts may undergo additional modification. The 5' cap may have several functions:

- to protect the transcript from 5'→3' exonuclease attack (uncapped mRNA molecules are rapidly degraded);
- to facilitate transport of mRNAs from the nucleus to the cytoplasm;
- to facilitate RNA splicing; and
- to facilitate attachment of the 40S subunit of cytoplasmic ribosomes to mRNA during translation.

3' polyadenylation

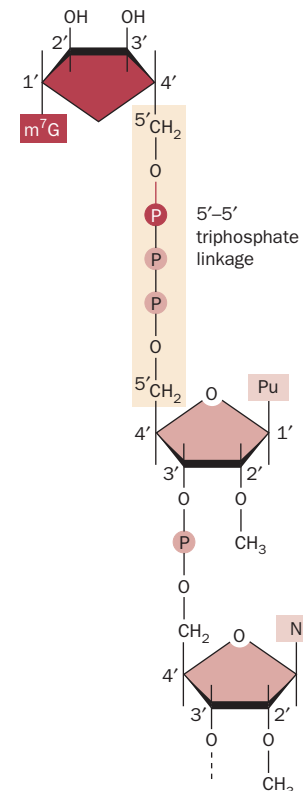
Transcription by both RNA polymerase I and III stops after the enzyme recognizes a specific transcription termination site. However, the 3' ends of mRNA molecules are determined by a post-transcriptional cleavage reaction. The sequence AAUAAA (sometimes AUUAAA) signals the 3' cleavage for most polymerase II transcripts.

Cleavage occurs at a specific site 15–30 nucleotides downstream of the AAUAAA sequence, although the primary transcript may continue for hundreds or even thousands of nucleotides past the cleavage point. After cleavage has occurred, the enzyme poly(A) polymerase sequentially adds adenylate (AMP) residues to the 3' end (about 200 in the case of mammalian mRNA). This polyadenylation reaction (Figure 1.21) produces a **poly(A) tail** that is thought to:

- Help transport mRNA to the cytoplasm.
- Stabilize at least some mRNA molecules in the cytoplasm.
- Enhance recognition of mRNA by the ribosomal machinery.

Histone genes are unique in producing mRNA that does not become polyadenylated; termination of their transcription nevertheless also involves 3' cleavage of the primary transcript.

Figure 1.20 The 5' cap of a eukaryotic mRNA. The nucleotide components in pink represent the residue of the original 5' end of a eukaryotic pre-mRNA. The primary pre-mRNA transcript begins with a nucleotide that contains a purine (Pu) base and a 5' triphosphate group. However, as the pre-mRNA undergoes processing, the end phosphate group at the 5' end is excised with a phosphatase to leave a 5' diphosphate group, and a specialized nucleotide is covalently joined to form a *cap* that will protect mRNA from exonuclease attack and assist in the initiation of translation. The cap nucleotide (with base shown in red) is first formed when a GTP residue is cleaved to generate a guanosine monophosphate that is then added through a 5'–5' triphosphate linkage (pale peach shading) to the diphosphate group of the original purine end nucleotide. Subsequently nitrogen atom 7 of the new 5' terminal G is methylated. In mRNAs synthesized in vertebrate cells, the 2' carbon atom of the ribose of each of the two adjacent nucleotides, the original purine end-nucleotide and its neighbor, are also methylated, as illustrated in this example. m⁷G, 7-methylguanosine; N, any nucleotide.



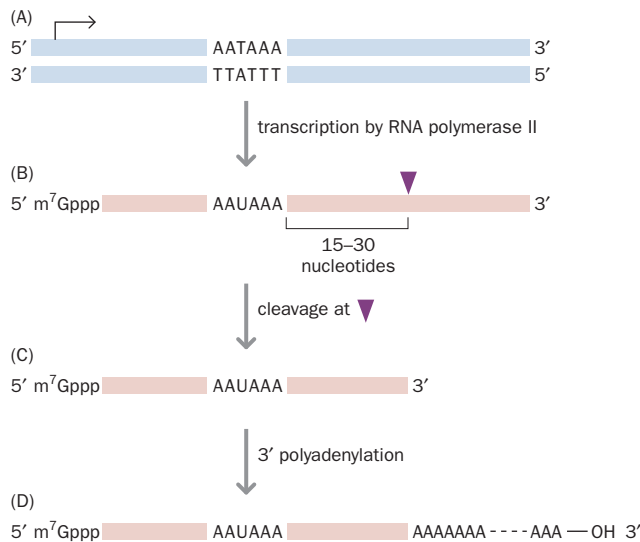


Figure 1.21 Polyadenylation of 3' ends of eukaryotic mRNAs. (A, B) As RNA polymerase II advances to transcribe a gene it carries at its rear two multiprotein complexes required for polyadenylation: CPSF (cleavage and polyadenylation specificity factor) and CStF (cleavage and stimulation factor) that cooperate to identify a polyadenylation signal downstream of the termination codon in the RNA transcript and to cut the transcript. The polyadenylation signal comprises an AAUAAA sequence or close variant and some poorly understood downstream signals. (C) Cleavage occurs normally about 15–30 nucleotides downstream of the AAUAAA element, and (D) AMP residues are subsequently added by poly(A) polymerase to form a poly(A) tail.

rRNA and tRNA transcripts undergo extensive processing

Four major classes of eukaryotic rRNA have been identified: 28S, 18S, 5.8S, and 5S rRNA (S is the Svedberg coefficient, a measure of how fast large molecular structures sediment in an ultracentrifuge, corresponding directly to size and shape). 18S rRNA is found in the small subunits of ribosomes; the other three are components of the large subunit. Very large amounts of rRNA are required for cells to perform protein synthesis, so many genes are devoted to making rRNA in the nucleolus, a visibly distinct compartment of the nucleus.

In human cells a cluster of approximately 250 genes synthesizes 5S rRNA using RNA polymerase III, which also transcribes some other small RNA species. The 28S, 18S, and 5.8S rRNAs are encoded by consecutive genes on a common 13 kb transcription unit (**Figure 1.22**) that is transcribed by RNA polymerase I. A compound unit of the 13 kb transcription unit and an adjacent 27 kb non-transcribed spacer is tandemly repeated about 30–40 times at the nucleolar organizer regions on the short arms of each of the five human acrocentric chromosomes (13, 14, 15, 21, and 22). These five clusters of rRNA genes, each about 1.5 million bases (Mb) long, are sometimes referred to as **ribosomal DNA (rDNA)**.

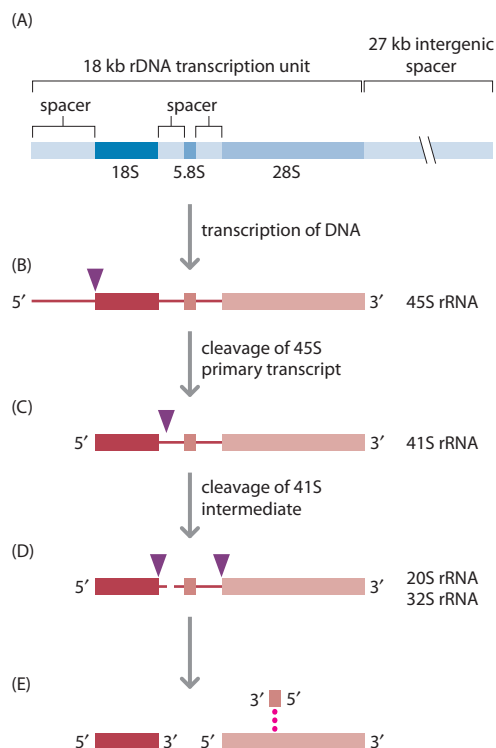


Figure 1.22 The major rRNA species are synthesized by cleavage of a shared primary transcript. (A) In human cells, the 18S, 5.8S, and 28S rRNAs are encoded by a single transcription unit that is 13 kb long. It occurs within tandem repeat units of about 40 kb that also includes a roughly 27 kb non-transcribed (intergenic) spacer. (B) Transcription by RNA polymerase I produces a 13 kb primary transcript (45S rRNA) that then undergoes a complex series of post-transcriptional cleavages. (C–E) Ultimately, individual 18S, 28S, and 5.8S rRNA molecules are released. The 18S rRNA will form part of the small ribosomal subunit. The 5.8S rRNA binds to a complementary segment of the 28S rRNA; the resulting complex will form part of the large ribosomal subunit. The latter also contains 5S rRNA, which is encoded separately by dedicated genes transcribed by RNA polymerase III.

In addition to the sequence of cleavage reactions (see Figure 1.22), the primary rRNA transcript also undergoes a variety of base-specific modifications. This extensive RNA processing is undertaken by many different small nucleolar RNAs that are encoded by about 200 different genes in the human genome.

Mature tRNA molecules also undergo extensive base modifications, and about 10% of the bases in any tRNA are modified versions of A, C, G, or U. Common examples of modified nucleosides include dihydrouridine, which has extra hydrogens at carbons 5 and 6; pseudouridine, an isomer of uridine; inosine (deaminated guanosine); and *N,N'*-dimethylguanosine.

1.5 TRANSLATION, POST-TRANSLATIONAL PROCESSING, AND PROTEIN STRUCTURE

The mRNA produced by genes in the nucleus migrates to the cytoplasm, where it engages with ribosomes and other components to initiate translation and polypeptide synthesis. Messenger RNA transcribed from genes in the mitochondria and chloroplasts is translated on dedicated ribosomes within these organelles.

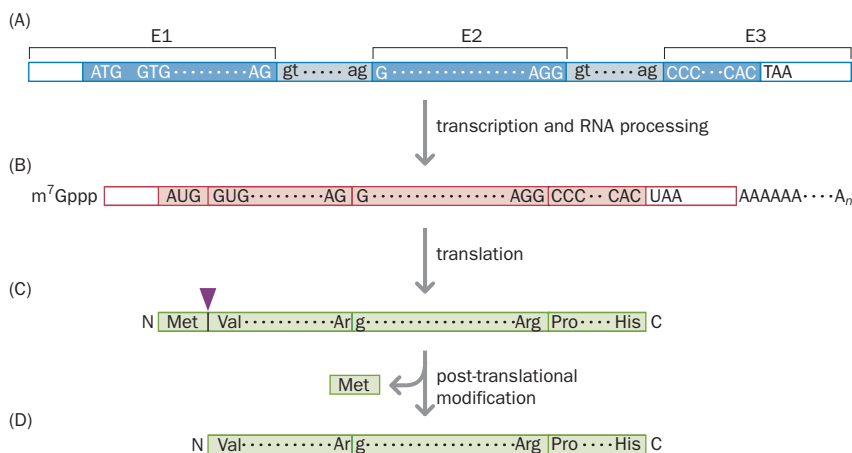
Only a central segment of a eukaryotic mRNA molecule is translated to make a polypeptide. The flanking **untranslated regions** (the 5' UTR and 3' UTR) are transcribed from exon sequences present at the 5' and 3' ends of the gene. They assist in binding and stabilizing the mRNA on the ribosomes, and promote efficient translation (Figure 1.23).

Ribosomes are large RNA–protein complexes composed of two subunits. In eukaryotes, cytoplasmic ribosomes have a large 60S subunit and a smaller 40S subunit. The 60S subunit contains three types of rRNA molecule: 28S rRNA, 5.8S rRNA, and 5S rRNA, as well as about 50 ribosomal proteins. The 40S subunit contains a single 18S rRNA and more than 30 ribosomal proteins. Ribosomes provide the structural framework for polypeptide synthesis. The RNA components are predominantly responsible for the catalytic function of the ribosome; the protein components are thought to enhance the function of the rRNA molecules, although a surprising number of them do not seem to be essential for ribosome function.

mRNA is decoded to specify polypeptides

The assembly of a new polypeptide from its constituent amino acids is governed by a triplet genetic code. Within an mRNA the central nucleotide sequence that is used to make polypeptide is scanned from 5' to 3' on the ribosome in groups of three nucleotides (**codons**). Each codon specifies an amino acid, and the decoding process uses a collection of different tRNA molecules, each of which binds one type of amino acid. An amino acid–tRNA complex is known as an aminoacyl tRNA and is formed when a dedicated aminoacyl tRNA synthetase covalently links the required amino acid to the terminal adenosine in the conserved CCA trinucleotide at the 3' end of the tRNA.

Figure 1.23 Transcription and translation of the human β -globin gene. (A) The β -globin gene comprises three exons (E1–E3) and two introns. The 5' end sequence of E1 and the 3' end sequence of E3 are noncoding sequences (unshaded sections). (B) These sequences are transcribed and so occur at the 5' and 3' ends (unshaded sections) of the β -globin mRNA that emerges from RNA processing. They are not, however, translated and so do not specify any part of the precursor polypeptide (C). This figure also illustrates that some codons can be specified by bases that are separated by an intron. The arginine at position 104 in the β -globin polypeptide is encoded by the last three nucleotides (AGG) of exon 2 but the arginine at position 30 is encoded by an AGG codon whose first two bases are encoded by the last two nucleotides of exon 1 and whose third base is encoded by the first nucleotide of exon 2. (D) During post-translational modification the 147-amino acid precursor polypeptide undergoes cleavage to remove its N-terminal methionine residue, to generate the mature 146-residue β -globin protein. The flanking N and C symbols to the left and right, respectively, in (C) and (D) depict the N-terminus (N) and C-terminus (C).



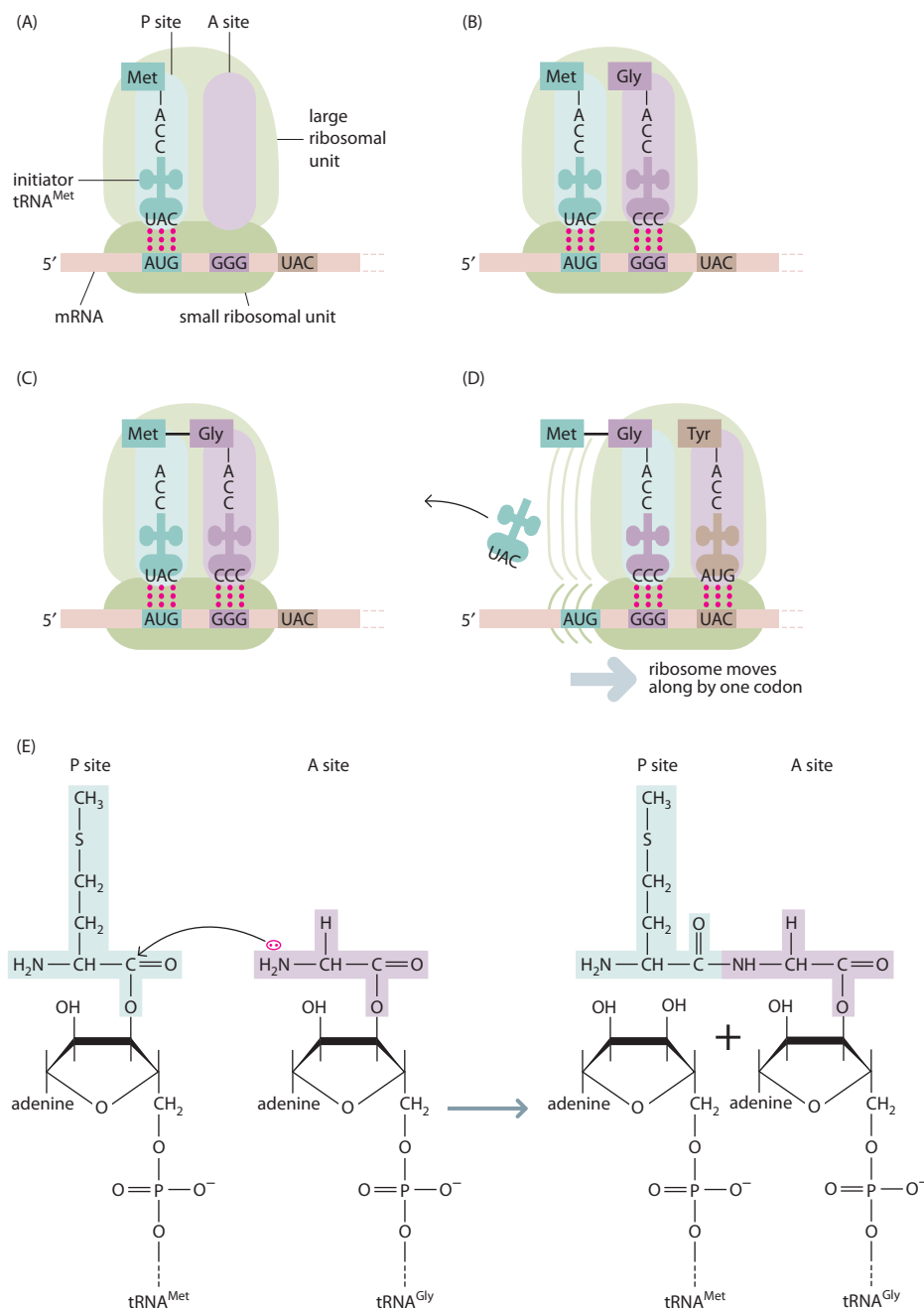


Figure 1.24 In translation, the genetic code is deciphered on ribosomes by **codon-anticodon recognition**. (A) The large ribosomal subunit (60S in eukaryotes) has two sites for binding an aminoacyl tRNA (a transfer RNA with its attached amino acid): the P (peptidyl) site and the A (aminoacyl) site. The small ribosomal subunit (40S in eukaryotes) binds mRNA, which is scanned along its 5' UTR in a 5'→3' direction until the start codon is identified, an AUG located within a larger consensus sequence (see the text). An initiator tRNA^{Met} carrying a methionine residue binds to the P site with its anticodon in register with the AUG start codon. (B) The appropriate aminoacyl tRNA is bound to the A site with its anticodon base-pairing with the next codon (GGG in this case, specifying glycine). (C) The rRNA in the large subunit catalyzes peptide bond formation, resulting in the methionine detaching from its tRNA and being bound instead to the glycine attached to the tRNA held at the A site. (D) The ribosome translocates along the mRNA so that the tRNA bearing the Met-Gly dipeptide is bound by the P site. The next aminoacyl tRNA (here, carrying Tyr) binds to the A site in preparation for new peptide bond formation. (E) Peptide bond formation. The N atom of the amino group of the amino acid bound to the tRNA in the A site makes a nucleophilic attack on the carboxyl C atom of the amino acid held by the tRNA bound to the P site.

Each tRNA has its own **anticodon**, a trinucleotide at the center of the anticodon arm (see Figure 1.9B) that provides the necessary specificity to interpret the genetic code. For an amino acid to be added to a growing polypeptide, the relevant codon of the mRNA molecule must be recognized by base pairing with a complementary anticodon on the appropriate aminoacyl tRNA molecule. This happens on the ribosome. The small ribosomal subunit binds the mRNA, and the large subunit has two sites for binding aminoacyl tRNAs, namely a P (peptidyl) site and an A (aminoacyl) site (Figure 1.24).

The cap at the 5' end of messenger RNA molecules is important in initiating translation. It is recognized by certain key proteins that bind the small ribosomal subunit, and these initiation factors hold the mRNA in place. In cap-dependent translation initiation, the ribosome scans the 5' UTR of the mRNA in the 5'→3' direction to find a suitable initiation codon, an AUG that is found within the Kozak consensus sequence 5'-GCCPuCCAUGG-3' (where Pu represents purine). The most important determinants are the G at position +4 (immediately following the AUG codon), and the purine (preferably A) at -3 (three nucleotides upstream of the AUG codon).

When a suitable initiation codon is identified, an initiating tRNA^{Met} with its attached methionine binds to the P site on the large ribosomal subunit so that its anticodon base-pairs with the AUG initiator codon on the mRNA (see Figure 1.24). Once this has happened, the transcriptional reading frame is established and codons are interpreted as successive groups of three nucleotides continuing in the 5'→3' direction downstream of the initiating AUG codon. An aminoacyl tRNA for the second codon (a tRNA^{Gly} to recognize GGG in the example of Figure 1.24) binds to the neighboring A site in the large subunit.

Once the P and A sites are occupied by aminoacyl tRNAs, the largest rRNA component within the large subunit of the ribosome is thought to act as a peptidyl transferase. It catalyzes the formation of a peptide bond by a condensation reaction between the amino group of the amino acid held by the tRNA in the A site and the carboxyl group of the methionine held by the tRNA^{Met}. The net result is to detach the initiator methionine from its tRNA and attach it to the second amino acid, forming a dipeptide (see Figure 1.24). Now without any attached amino acid, the tRNA^{Met} migrates away from the P site and its place is taken by the tRNA with the attached dipeptide that formerly occupied the A site. The liberated A site is now filled by an aminoacyl tRNA carrying an anticodon that is complementary to the third codon, and a new peptide bond is formed to make a tripeptide, and so on.

After a ribosome has initiated translation of an mRNA and has then moved along the mRNA, other ribosomes can engage with the same mRNA. The resulting polyribosome structures (polysomes) make multiple copies of a polypeptide from the one mRNA molecule. Polypeptide chain elongation occurs until a termination codon is met. For mRNA transcribed from nuclear genes, termination codons come in three varieties: UAA (ochre), UAG (amber), and UGA (opal), but there are some differences for mitochondrial mRNA as described in the next section.

In response to a termination codon a protein release factor enters the A site instead of an aminoacyl tRNA to signal that the polypeptide should disengage from the ribosome. The completed polypeptide will then undergo processing that can include cleavage and modification of the side chains. Its backbone will have a free amino group at one end (the N-terminal end) and a free carboxyl group at the other end (the C-terminal end).

The genetic code is degenerate and not quite universal

The genetic code is a three-letter code, and there are four possible bases to choose from at each of the three base positions in a codon. There are therefore $4^3 = 64$ possible codons, which is more than sufficient to encode the 20 major types of amino acid. The genetic code is degenerate because, on average, each amino acid is specified by about three different codons. Some amino acids (such as leucine, serine, and arginine) are specified by as many as six codons; others are much more poorly represented (Figure 1.25). The degeneracy of the genetic code most often involves the third base of the codon.

mtDNA variants			mtDNA variant		
AAA	Lys	CAA	Gln	GAA	Glu
AAG		CAG		GAG	
AAC	Asn	CAC	His	GAC	Asp
AAU		CAU		GAU	
ACA		CCA	Pro	GCA	
ACG	Thr	CCG		GCG	Ala
ACC		CCC		GCC	
ACU		CCU		GCU	
UAA	STOP	CGA	Arg	GGA	
UAG		CGG		GGG	Gly
UAC		CGC	Arg	GGC	
UAU		CGU		GGU	
UUA	Leu	CUA		GUA	
UUG		CUG	Leu	GUG	Val
UUC		CUC		GUC	
UUU	Phe	CUU		GUU	

Figure 1.25 The genetic code. All 64 possible codons of the genetic code and the amino acid specified by each, as read in the 5'→3' direction from the mRNA sequence. The interpretations of the 64 codons in the 'universal' genetic code are shown in black immediately to the right of the codons. Sixty-one codons specify an amino acid. Three STOP codons (UAA, UAG, and UGA) do not encode any amino acid. The genetic code for mitochondrial mRNA (mtDNA) conforms to the universal code except for a few variants. For example, in the mitochondrial genetic code in humans and many other species four codons are used differently: UGA encodes tryptophan instead of being a STOP codon, AUA encodes methionine, and instead of encoding arginine, AGA and AGG are STOP codons.

Although more than 60 codons can specify an amino acid, the number of different cytoplasmic tRNA molecules is quite a bit smaller, and only 22 types of mitochondrial tRNA are made. The interpretation of more than 60 sense codons with a much smaller number of different tRNAs is possible because base pairing in RNA is more flexible than in DNA. Pairing of codon and anticodon follows the normal A–U and G–C rules for the first two base positions in a codon. However, at the third position there is some flexibility (base wobble), and GU base pairs are tolerated here (Table 1.6).

The genetic code is the same throughout nearly all life forms. However, mitochondria and chloroplasts have a limited capacity for protein synthesis, and during evolution their genetic codes have diverged slightly from that used at cytoplasmic ribosomes. Translation of nuclear-encoded mRNA continues until one of three stop codons is encountered (UAA, UAG, or UGA) but in mammalian mitochondria there are four possibilities (UAA, UAG, AGA, and AGG).

The meaning of a codon can also be dependent upon the sequence context; that is, the nature of the nucleotide sequence in which it is embedded. Depending on the surrounding sequence, some codons in a few types of nuclear-encoded mRNA can be interpreted differently from normal. For example, in a wide variety of cells the stop codon UGA is alternatively interpreted as encoding selenocysteine with some nuclear-encoded mRNAs, and UAG can sometimes be interpreted to encode glutamine.

TABLE 1.6 RULES FOR BASE PAIRING CAN BE RELAXED (WOBBLE) AT POSITION 3 OF A CODON

Base at 5' end of tRNA anticodon	Base recognized at 3' end of mRNA codon
A	U only
C	G only
G (or I) ^a	C or U
U	A or G

^aInosine (I) is a deaminated form of guanosine.

Post-translational processing: chemical modification of amino acids and polypeptide cleavage

Primary translation products often undergo a variety of modifications during or after translation. Simple or complex chemical groups are often covalently attached to the side chains of certain amino acids (Table 1.7). In addition, polypeptides may occasionally be cleaved to yield one or more active polypeptide products.

Addition of carbohydrate groups

Glycoproteins have oligosaccharides covalently attached to the side chains of certain amino acids. Few proteins in the cytosol are glycosylated (carry an

TABLE 1.7 MAJOR TYPES OF MODIFICATION OF POLYPEPTIDES

Type of modification (group added)	Target amino acid(s)	Notes
Phosphorylation (PO ₄ ⁻)	Tyr, Ser, Thr	achieved by specific kinases; may be reversed by phosphatases
Methylation (CH ₃)	Lys	achieved by methylases; reversed by demethylases
Hydroxylation (OH)	Pro, Lys, Asp	hydroxyproline (Hyp) and hydroxylysine (Hyl) are particularly common in collagens
Acetylation (CH ₃ CO)	Lys	achieved by an acetylase; reversed by deacetylase
Carboxylation (COOH)	Glu	achieved by γ-carboxylase
N-glycosylation (complex carbohydrate)	Asn ^a	takes place initially in the endoplasmic reticulum, with later additional changes occurring in the Golgi apparatus
O-glycosylation (complex carbohydrate)	Ser, Thr, Hyl ^b	takes place in the Golgi apparatus; less common than N-glycosylation
Glycosylphosphatidylinositol (glycolipid)	Asp ^c	serves to anchor protein to outer layer of plasma membrane
Myristoylation (C ₁₄ fatty acyl group)	Gly ^d	serves as membrane anchor
Palmitoylation (C ₁₆ fatty acyl group)	Cys ^e	serves as membrane anchor
Farnesylation (C ₁₅ prenyl group)	Cys ^c	serves as membrane anchor
Geranylgeranylation (C ₂₀ prenyl group)	Cys ^c	serves as membrane anchor

^aThis is especially common when Asn is in the sequence: Asn-X-(Ser/Thr), where X is any amino acid other than Pro. ^bHydroxylysine. ^cAt C-terminus of polypeptide. ^dAt N-terminus of polypeptide. ^eTo form S-palmitoyl link.

attached carbohydrate); if they are, they have a single sugar residue, *N*-acetylglucosamine, attached to a serine or threonine residue. However, proteins that are secreted from cells or transported to lysosomes, the Golgi apparatus, or the plasma membrane are routinely glycosylated. In these cases, the sugars are assembled as oligosaccharides before being attached to the protein.

Two major types of glycosylation occur. Carbohydrate *N*-glycosylation involves attaching a carbohydrate group to the nitrogen atom of an asparagine side chain, and *O*-glycosylation entails adding a carbohydrate to the oxygen atom of an OH group carried by the side chains of certain amino acids (see Table 1.7).

Proteoglycans are proteins with attached glycosaminoglycans (polysaccharides) that usually include repeating disaccharide units containing glucosamine or galactosamine. The best-characterized proteoglycans are components of the extracellular matrix, a complex network of macromolecules secreted by, and surrounding, cells in tissues or in culture systems.

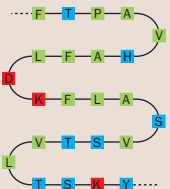
Addition of lipid groups

Some proteins, notably membrane proteins, are modified by the addition of fatty acyl or prenyl groups. These added groups typically serve as membrane anchors, hydrophobic amino acid sequences that secure a newly synthesized protein within either a plasma membrane or the endoplasmic reticulum (Table 1.8).

Anchoring a protein to the outer layer of the plasma membrane involves the attachment of a glycosylphosphatidylinositol (GPI) group. This glycolipid group contains a fatty acyl group that serves as the membrane anchor; it is linked successively to a glycerophosphate unit, an oligosaccharide unit, and finally—through a phosphoethanolamine unit—to the C-terminus of the protein. The entire protein, except the GPI anchor, is located in the extracellular space.

Post-translational cleavage

The primary translation product may also undergo internal cleavage to generate a smaller mature product. Occasionally the initiating methionine is cleaved from

TABLE 1.8 LEVELS OF PROTEIN STRUCTURE		
Level	Definition	Notes
Primary 	the linear sequence of amino acids in a polypeptide	can vary enormously in length from a few to thousands of amino acids
Secondary 	the path that a polypeptide backbone follows within local regions of the primary structure	varies along the length of the polypeptide; common elements of secondary structure include the α -helix and β -pleated sheet
Tertiary 	the overall three-dimensional structure of a polypeptide, arising from the combination of all of the secondary structures	can take various forms (e.g. globular, rod-like, tube, coil, sheet)
Quaternary 	the aggregate structure of a multimeric protein (comprising more than one subunit, which may be of more than one type)	can be stabilized by disulfide bridges between subunits or ligand binding, and other factors

the primary translation product, as during the synthesis of β -globin (see Figure 1.23C, D). More substantial polypeptide cleavage is observed during the maturation of many proteins, including plasma proteins, polypeptide hormones, neuropeptides, and growth factors. Cleavable signal sequences are often used to mark proteins either for export or for transport to a specific intracellular location. A single mRNA molecule can sometimes specify more than one functional polypeptide chain as a result of post-translational cleavage of a large precursor polypeptide (Figure 1.26).

The complex relationship between amino acid sequence and protein structure

Proteins can be composed of one or more polypeptides, each of which may be subject to post-translational modification. Interactions between a protein and either of the following may substantially alter the conformation of that protein:

- A **cofactor**, such as a divalent cation (such as Ca^{2+} , Fe^{2+} , Cu^{2+} , or Zn^{2+}), or a small molecule required for functional enzyme activity (such as NAD^+).
- A **ligand** (any molecule that a protein binds specifically).

Four different levels of structural organization in proteins have been distinguished and defined (see Table 1.8).

Even within a single polypeptide, there is ample scope for hydrogen bonding between different amino acid residues. This stabilizes the partial polar charges along the backbone of the polypeptide and has profound effects on that protein's overall shape. With regard to a protein's conformation, the most significant hydrogen bonds are those that occur between the oxygen of one peptide bond's carbonyl ($\text{C}=\text{O}$) group and the hydrogen of the amino (NH) group of another peptide bond. Several fundamental structural patterns (motifs) stabilized by hydrogen bonding within a single polypeptide have been identified, the most fundamental of which are described below.

The α -helix

This is a rigid cylinder that is stabilized by hydrogen bonding between the carbonyl oxygen of a peptide bond and the hydrogen atom of the amino nitrogen of a peptide bond located four amino acids away (Figure 1.27). α -Helices often occur

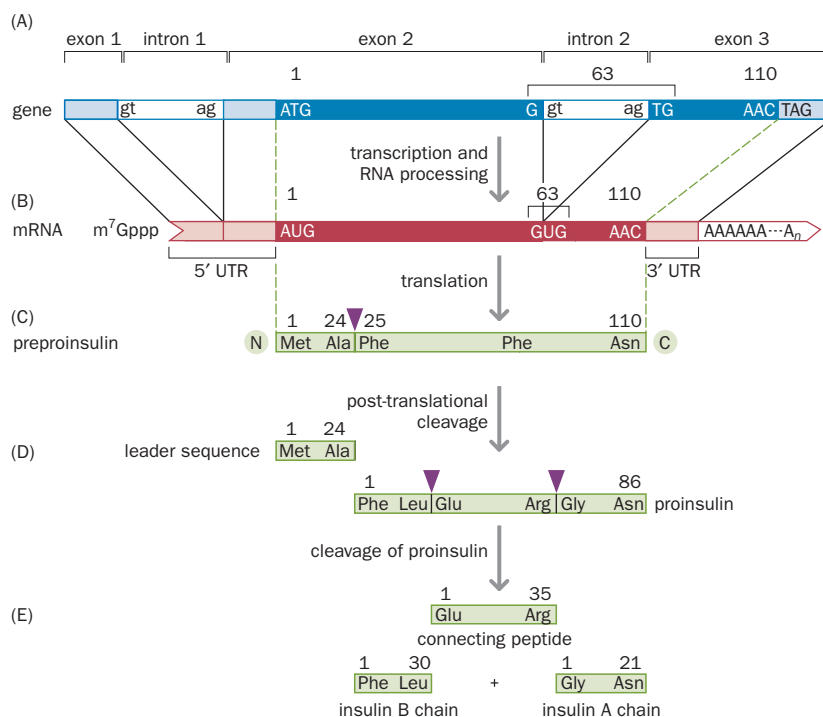


Figure 1.26 Insulin synthesis involves multiple post-translational cleavages of polypeptide precursors. (A) The human insulin gene comprises three exons and two introns. The coding sequence (the part that will be used to make polypeptide) is shown in deep blue. It is confined to the 3' sequence of exon 2 and the 5' sequence of exon 3. (B) Exon 1 and the 5' part of exon 2 specify the 5' untranslated region (5' UTR), and the 3' end of exon 3 specifies the 3' UTR. The UTRs are transcribed and so are present at the ends of the mRNA. (C) A primary translation product, preproinsulin, has 110 residues and is cleaved to give (D) a 24-residue N-terminal *leader sequence* (that is required for the protein to cross the cell membrane but is thereafter discarded) plus an 86-residue proinsulin precursor. (E) Proinsulin is cleaved to give a central segment (the connecting peptide) that may maintain the conformation of the A and B chains of insulin before the formation of their interconnecting covalent disulfide bridges (see Figure 1.29).

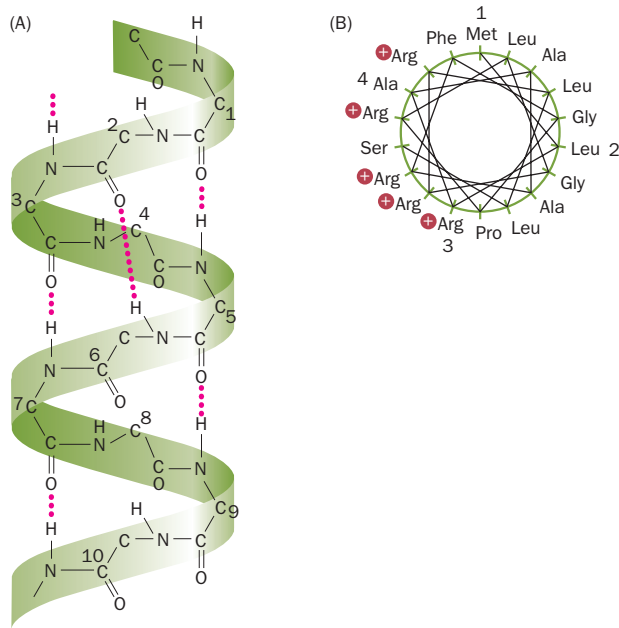


Figure 1.27 The structure of a standard α -helix and an amphipathic α -helix.

(A) The structure of an α -helix is stabilized by hydrogen bonding between the oxygen of the carbonyl group (C=O) of each peptide bond and the hydrogen on the peptide bond amide group (NH) of the fourth amino acid away, making the helix have 3.6 amino acid residues per turn. The side chains of each amino acid are located on the outside of the helix; there is almost no free space within the helix. Note that only the backbone of the polypeptide is shown, and some bonds have been omitted for clarity. (B) An amphipathic α -helix has tighter packing and has charged amino acids and hydrophobic amino acids located on different surfaces. Here we show an end view of such a helix: five positively charged arginine residues are clustered on one side of the helix, whereas the opposing side has a series of hydrophobic amino acids (mostly Ala, Leu, and Gly). The lines within the circle indicate neighboring residues—the initiator methionine (position 1) is connected to a leucine (2), which is connected to an arginine (3), which is adjacent to an alanine (4), and so on.

in proteins that perform key cellular functions (such as transcription factors, where they are usually represented in the DNA-binding domains). Identical α -helices with a repeating arrangement of nonpolar side chains can coil round each other to form a particularly stable coiled coil. Coiled coils occur in many fibrous proteins, such as collagen of the extracellular matrix, the muscle protein tropomyosin, α -keratin in hair, and fibrinogen in blood clots.

The β -pleated sheet

β -Pleated sheets are also stabilized by hydrogen bonding but, in this case, they occur between opposed peptide bonds in parallel or anti-parallel segments of the same polypeptide chain (Figure 1.28). β -Pleated sheets occur—often together with α -helices—at the core of most globular proteins.

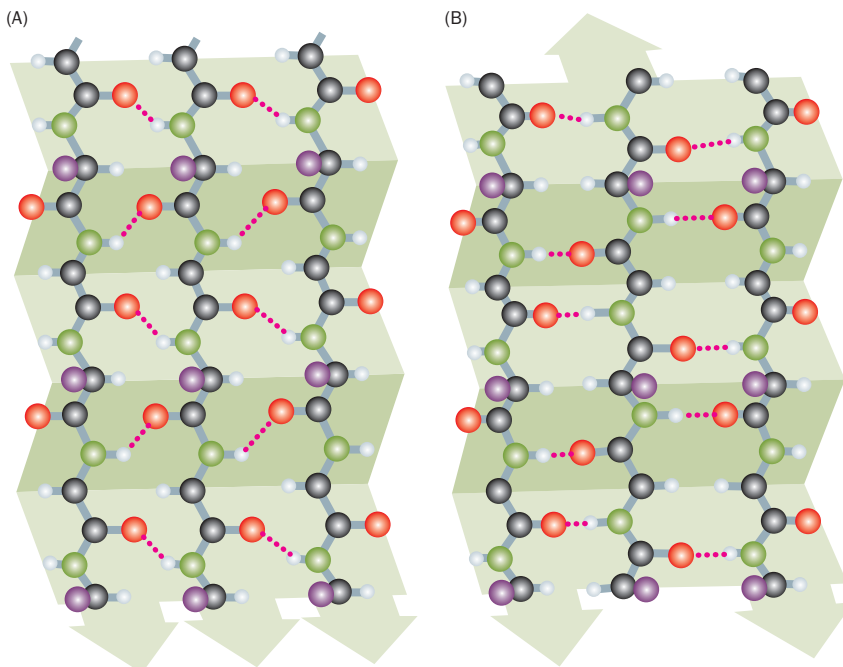


Figure 1.28 The structure of a β -pleated sheet. Hydrogen bonding occurs here between the carbonyl (C=O) oxygens and amide (NH) hydrogens on adjacent segments of (A) parallel and (B) anti-parallel β -pleated sheets. [Adapted from Lehninger AL, Nelson DL & Cox MM (1993) Principles of Biochemistry, 2nd ed. With permission from WH Freeman and Company.]

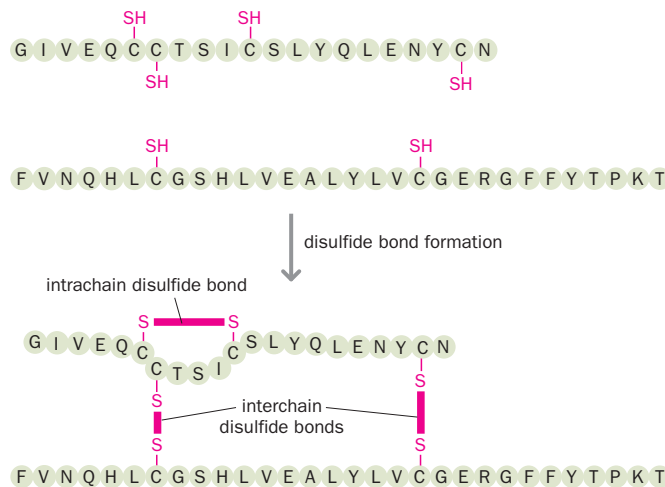


Figure 1.29 Intrachain and interchain disulfide bridges in human insulin. Disulfide bridges ($-S-S-$) form by a condensation reaction between the sulfhydryl ($-SH$) groups on the side chains of cysteine residues. They can form between cysteine side chains within the same polypeptide (such as between positions 6 and 11 within the insulin A chain) and also between cysteine side chains on different interacting polypeptides such as the insulin A and B chains.

The β -turn

Hydrogen bonding can occur between amino acids that are even nearer to each other within a polypeptide. When this arises between the peptide bond CO group of one amino acid residue and the peptide bond NH group of an amino acid residue three places farther along, this results in a hairpin β -turn. Abrupt changes in the direction of a polypeptide enable compact globular shapes to be achieved. These β -turns can connect parallel or anti-parallel strands in β -pleated sheets.

Higher-order structures

Many more complex structural motifs, consisting of combinations of the above structural modules, form **protein domains**. Such domains are often crucial to a protein's overall shape and stability and usually represent functional units involved in binding other molecules. Another important determinant of the structure (and function) of a protein is **disulfide bridges**. They can form between the sulfur atoms of sulfhydryl ($-SH$) groups on two amino acids that may reside on a single polypeptide chain or on two polypeptide chains (**Figure 1.29**).

In general, the primary structure of a protein determines the set of secondary structures that, together, generate the protein's tertiary structure. Secondary structural motifs can be predicted from an analysis of the primary structure, but the overall tertiary structure cannot easily be accurately predicted. Finally, some proteins form complex aggregates of polypeptide subunits, giving an arrangement known as the quaternary structure.

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Chromosome Structure and Function

2

KEY CONCEPTS

- Chromosomes have two fundamental roles: the faithful transmission and appropriate expression of genetic information.
- Prokaryotic chromosomes contain circular double-stranded DNA molecules that are relatively protein-free, but eukaryotic chromosomes consist of linear double-stranded DNA molecules complexed throughout their lengths with proteins.
- Chromatin is the DNA-protein matrix of eukaryotic chromosomes. The complexed proteins serve structural roles, including compacting the DNA in different ways, and also regulatory roles.
- Chromosomes undergo major changes in the cell cycle, notably at S phase when they replicate and at M phase when the replicated chromosomes become separated and allocated to two daughter cells.
- DNA replication at S phase produces two double-stranded daughter DNA molecules that are held together at a specialized region, the centromere. When the daughter DNA molecules remain held together like this they are known as sister chromatids, but once they separate at M phase they become individual chromosomes.
- At the metaphase stage of M phase the chromosomes are so highly condensed that gene expression is uniformly shut down. But this is the optimal time for viewing them under the microscope. Staining with dyes that bind preferentially to GC-rich or AT-rich regions can give reproducible chromosome banding patterns that allow different chromosomes to be differentiated.
- During interphase, the long period of the cell cycle that separates successive M phases, chromosomes have generally very long extended conformations and are invisible under optical microscopy. The extended structure means that genes can be expressed efficiently.
- Even during interphase some chromosomal regions always remain highly condensed and transcriptionally inactive (heterochromatin), whereas others are extended to allow gene expression (euchromatin).
- Sperm and egg cells have one copy of each chromosome (they are haploid), but most cells are diploid, having two sets of chromosomes.
- Fertilization of a haploid egg by a haploid sperm generates the diploid zygote from which all other body cells arise by cell division.
- In mitosis a cell divides to give two daughter cells, each with the same number and types of chromosomes as the original cell.
- Meiosis is a specialized form of cell division that occurs in certain cells of the testes and ovaries to produce haploid sperm and egg cells. During meiosis new genetic combinations are randomly created, partly by exchanging sequences between maternal and paternal chromosomes.
- Three types of functional element are needed for eukaryotic chromosomes to transmit DNA faithfully from mother cell to daughter cells: the centromere (ensures correct chromosome segregation at cell division); replication origins (initiate DNA replication); and telomeres (cap the chromosomes to stop the internal DNA from being degraded by nucleases).
- An abnormal number of chromosomes can sometimes occur but this is often lethal if present in most cells of the body.
- Structural chromosome abnormalities arising from breaks in chromosomes can cause genes to be deleted or incorrectly expressed.
- Having the correct number and structure of chromosomes is not enough. They must also have the correct parental origin because certain genes are preferentially expressed on either paternally or maternally inherited chromosomes.

The underlying structure and fundamental functions of DNA—replication and transcription—were introduced in the previous chapter. But DNA functions in a context. In eukaryotic cells, the very long DNA molecules in the nucleus are complexed with a variety of structural and regulatory proteins and structured into linear chromosomes. The DNA molecules in mitochondria are different: they are comparatively short, have little protein attached to them, and are circular.

This chapter introduces the life cycle of chromosomes in eukaryotic cells that are usually formed from other cells by cell division. The process of cell division is a small component of the cell cycle, the process in which chromosomes and their constituent DNA molecules need to make perfect copies of themselves and then segregate into daughter cells. There are important differences between how this occurs in routine cell division and in the specialized form of cell division that gives rise to sperm and egg cells.

A feature common to both types of cell division is the importance to the cell of chromosome condensation. This affects the expression of information encoded in the DNA and makes long and fragile DNA strands resilient to breakage during the dramatic rearrangements that occur in cell division. The use of dyes to stain condensed chromosomes has revealed patterns that, like fingerprints, can be used to distinguish between them. Careful examination of these and other patterns can reveal evidence of chromosomal abnormalities, such as breakages and rearrangements that have occurred and survived but may cause disease.

2.1 PLOIDY AND THE CELL CYCLE

The chromosome and DNA content of cells is defined by the number (n) of different chromosomes, the **chromosome set**, and its associated DNA content (C). For human cells, $n = 23$ and C is about 3.5 pg (3.5×10^{-12} g). Different cell types in an organism, however, may differ in **ploidy**—the number of copies they have of the chromosome set. Sperm and egg cells carry a single chromosome set and are said to be **haploid** (they have n chromosomes and a DNA content of C). Most human and mammalian cells carry two copies of the chromosome set and are **diploid** (having $2n$ chromosomes and a DNA content of $2C$). However, in several non-mammalian animal species most of the body cells are not diploid but instead are either haploid or polyploid. In the latter case, some species are tetraploid ($4n$) and others have a ploidy of more than $4n$, but triploidy ($3n$) is less common in animals because it can give rise to problems in producing sperm and egg cells.

The cells of our body are all derived ultimately from a single diploid cell, the **zygote**, that is formed when a sperm fertilizes an egg. Starting from the zygote, organisms grow by repeated rounds of cell division. Each round of cell division is a **cell cycle** and comprises a brief M phase, during which cell division occurs, and the much longer intervening **interphase**, which has three parts (Figure 2.1). They are: S phase (during which DNA synthesis occurs), G_1 phase (the gap between M phase and S phase), and G_2 phase (the gap between S phase and M phase).

We will describe the cell biology underlying the phases of the cell cycle in a later chapter. Here we are concerned with the life cycle of chromosomes. During each cell cycle, chromosomes undergo profound changes to their structure, number, and distribution within the cell. From the end of M phase right through until DNA duplication in S phase, a chromosome of a diploid cell contains a single DNA double helix and the total DNA content is $2C$ (see Figure 2.1). After DNA duplication, the total DNA content is $4C$, but the duplicated double helices are held together along their lengths so that each chromosome has double the DNA content of a chromosome in early S phase. During M phase the duplicated double helices separate, generating two daughter chromosomes, giving $4n$ chromosomes. After equal distribution of the chromosomes to the two daughter cells, both cells will have $2n$ chromosomes and a DNA content of $2C$ (see Figure 2.1).

G_1 is the normal state of a cell, and is the long-term end state of non-dividing cells. Cells enter S phase only if they are committed to mitosis; as will be described in more detail in Chapter 4, non-dividing cells remain in a modified G_1 stage, sometimes called the G_0 phase. The cell cycle diagram can give the impression that all the interesting action happens in S and M phases—but this is an illusion. A cell spends most of its life in G_0 or G_1 phase, and that is where the genome does most of its work.

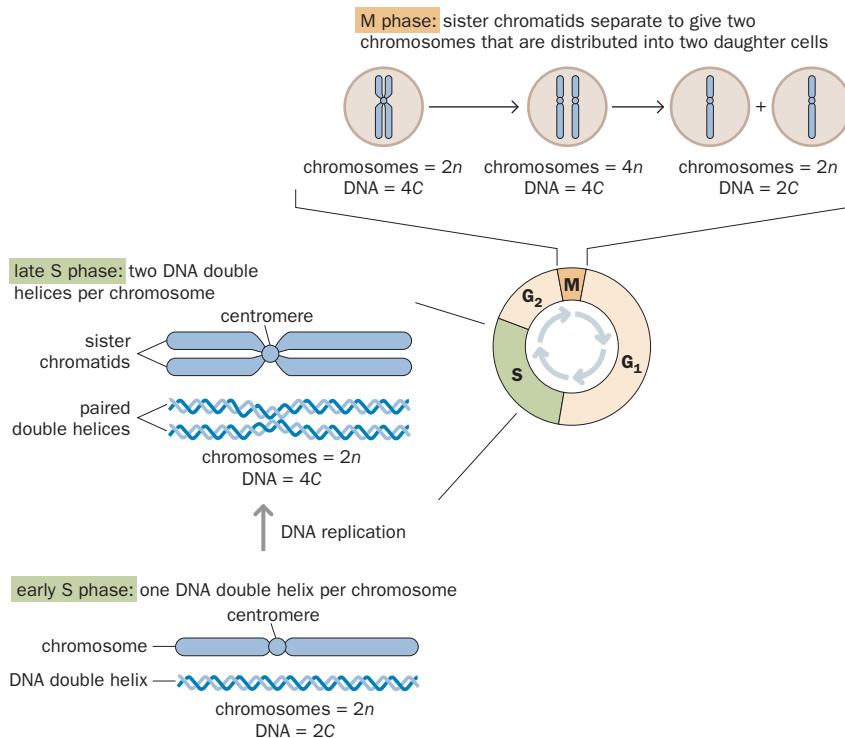


Figure 2.1 Changes in chromosomes and DNA content during the cell cycle. The cell cycle shown at the right includes a very short M phase, when the chromosomes become extremely highly condensed in preparation for nuclear and cell division. Afterwards, cells enter a long period of growth called interphase, during which chromosomes are enormously extended so that genes can be expressed. Interphase is divided into three phases: G₁, S (when the DNA replicates), and G₂. Chromosomes contain one DNA double helix from the end of M phase right through until just before the DNA is duplicated in S phase. After the DNA double helix has been duplicated, the two resulting double helices are held together tightly along their lengths (by specialized protein complexes called cohesins) until M phase. As the chromosomes condense at M phase they are now seen to consist of two sister chromatids, each containing a DNA duplex, that are bound together only at the centromeres. During M phase the two sister chromatids separate to form two independent chromosomes that are then equally distributed into the daughter cells.

A small subset of diploid body cells constitute the **germ line** that gives rise to **gametes** (sperm cells or egg cells). In humans, where $n = 23$, each gamete contains one sex chromosome plus 22 non-sex chromosomes (**autosomes**). In eggs, the sex chromosome is always an X; in sperm it may be either an X or a Y. After a haploid sperm fertilizes a haploid egg, the resulting diploid zygote and almost all of its descendant cells have the chromosome constitution 46,XX (female) or 46,XY (male) (**Figure 2.2**).

Cells outside the germ line are **somatic cells**. Human somatic cells are usually diploid but, as will be described later, there are notable exceptions. Some types of non-dividing cell lack a nucleus and any chromosomes, and so are nulliploid. Other cell types have multiple chromosome sets; they are naturally polyploid as a result of multiple rounds of DNA replication without cell division.

2.2 MITOSIS AND MEIOSIS

Mitosis and meiosis are both cell division processes that involve chromosome replication and cell division. However, the products of mitosis have the same ploidy as the initiating cell, whereas meiosis halves the cell's ploidy. Furthermore, whereas mitosis gives rise to genetically identical products, meiosis generates genetic diversity to ensure that offspring are genetically different from their parents.

Mitosis is the normal form of cell division

As an embryo develops through fetus, infant, and child to adult, many cell cycles are needed to generate the required number of cells. Because many cells have a limited life span, there is also a continuous requirement to generate new cells,

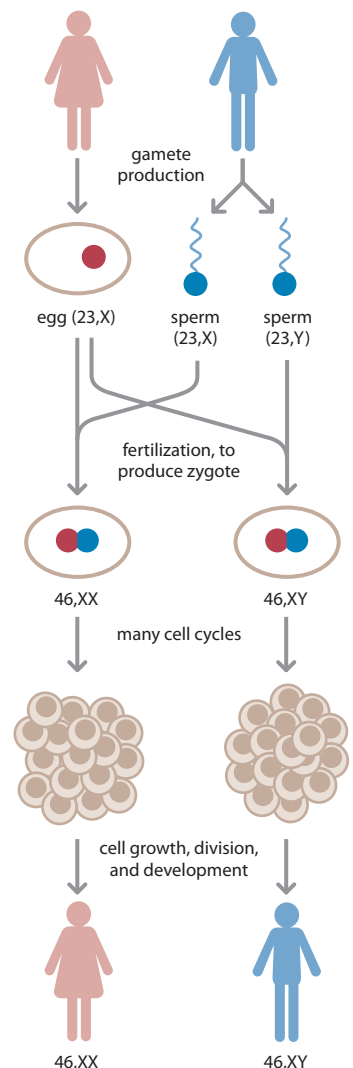


Figure 2.2 The human life cycle, from a chromosomal viewpoint. Haploid egg and sperm cells originate from diploid precursors in the ovary and testis in women and men, respectively. All eggs have a 23,X chromosome constitution, representing 22 autosomes plus a single X sex chromosome. A sperm can carry either sex chromosome, so that the chromosome constitution is 23,X (50%) and 23,Y (50%). After fertilization and fusion of the egg and sperm nuclei, the diploid zygote will have a chromosome constitution of either 46,XX or 46,XY, depending on which sex chromosome the fertilizing sperm carried. After many cell cycles, this zygote gives rise to all cells of the adult body, almost all of which will have the same chromosome complement as the zygote from which they originated.